



Effect of land-use change and climate change on the structure and diversity of woody vegetation: lessons from West Africa's woodlands

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Abstract

Land-use change in terrestrial ecosystems play a crucial role in many global change processes and has a considerable effect on biodiversity and ecosystem functioning. In addition to changes in how land is being used, human-induced climate change in the last centuries has substantially influenced global environmental patterns. Yet, our understanding of how the combined impact of these global change processes will affect ecosystems are still limited particularly in West Africa, where increasing human pressures and climatic change patterns are prominent. In this study, we therefore examined the effects of these interactions (between climate change, and land-use change) on the structure and diversity of woody vegetation in West Africa's woodlands. Here, we took advantage of the steep regional gradient of climatic aridity (Humid, Semi-arid) and local gradients of land-use intensification (Forest, Fallow, and Cropland) that exist in West Africa and applied a crossed-space-for-time substitution to capture the isolated and interactive effects of climate change and land-use change. Inventories of woody vegetation were conducted on 120 plots randomly laid out along these varying levels of land-use intensification and climate aridity in Ghana. From the result of an extensive field data, we demonstrated with our study that land-use intensification significantly impacts the taxonomic diversity of woody vegetation in West Africa as we recorded the lowest species diversity in Cropland (highest degree of intensity and disturbance), while Forest (protected areas) held the highest species diversity values. Similarly, with respect to the structure of woody vegetation, we compared population densities, the performance of juvenile and adult age-classes, and further analysed size-class distributions of each land-use type. Our study demonstrated that with increasing land-use pressure, the health of tree population reduces. Compared to other land-use types, proportions of Juveniles on Croplands were the lowest, leading to an almost flat slope with very little to low recruitment. The importance of our study implies that sustainable land management practices could potentially mitigate the detrimental effects of global ecological change on taxonomic diversity and health of woody populations in West Africa.

Declaration

I, Maximus Tochukwu Anochirim, hereby declare to the Senate of the University of Greifswald that this Master's thesis titled "Effect of land-use change and climate change on the structure and diversity of woody vegetation: lessons from West Africa's woodlands" and the work presented in the thesis are my original work. I confirm that the thesis has not been published or submitted to any other university for the purpose of obtaining any other diploma or degree.

Maximus Tochukwu Anochirim
(MSc. candidate)

Place, Date

Dedication

This thesis is dedicated to my parents (Michael and Genevieve Anochirim), who first planted and nourished the seeds of Environmental stewardship within me. It is further dedicated to my fellow stewards (Chiamaka, Nnaemeka, Obioma and Oluchi), to my community of friends and to everyone else working towards making the world a better place to live in.

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1. INTRODUCTION

Globally, woodlands play a crucial role in the proper functioning of ecosystems and in the provision of important services including the storage of carbon, regulation of the Earth's climate, as a source of food and water, and in supporting biodiversity (Millennium Ecosystem Assessment, 2005). Likewise, woodlands provide numerous ecological, cultural, and socio-economic benefits to local communities including acting as valuable grazing lands, and the provision of timber and non-timber forest products (Schumann et al. 2011; Linstädter et al. 2013; Sinare and Gordon, 2015). As a result of the high dependence of local communities on the vital resources these ecosystems provide, woodlands are highly prone to anthropogenic disturbances and can subsequently serve as excellent models to understand the impact of disturbances over space and time.

In West Africa, woodlands have diverse vegetation formations and are characterized by a mosaic of savannas, open grasslands and forest (Chidumayo and Gumbo, 2010). As a result, this biome not only supports a high diversity of woody plant species (many of which are endemic to the region), they also provide valuable ecological, economic (Atanasso et al. 2019), and cultural values (Sop et al. 2012; Ouédraogo et al. 2014). However, since the early nineties, there have been concerns about declining woody vegetation cover in West Africa (Lykke et al. 1998a; 1998b; 1999). In addition, significant changes in the composition, diversity, and structure of West African woodlands have also been documented (McClean et al. 2005; Brink and Eva, 2009; Schmidt et al. 2010; Heubes et al. 2013; Sule and Odekunle, 2016). In an effort to understand the impact of these documented losses on ecosystem health, develop appropriate responses and more specifically determine the drivers of these changes, a multitude of studies (e.g. Vitousek, 1992; 1994; Sala et al. 2000; Hawkins et al. 2003; Ross et al. 2021) were established; thus laying the foundation for an active research field of global change ecology. Among the global environmental change drivers, land-use change coupled with human-induced climate change were found to have the most impact on biodiversity loss (Vitousek, 1992; Sala et al. 2000; Wittig et al. 2007) with strong effects on ecosystem functions and services (Sala et al. 2000). Particularly in terrestrial ecosystems, land-use plays a vital role in many global change processes (Alcamo et al. 2011) and consequently could have a negative effect on biodiversity and the functioning of ecosystems (Sala et al. 2000; Díaz et al. 2007; Pfeifer et al. 2017). In West Africa, land-use change and forestry - LUCF is also among the leading sources of emissions from greenhouse gases (GHG) (IPCC, 2014a). Concerns about the pace and extent of land-use and land cover change are therefore very high on the global change research agenda, particularly with the realization of the important role of these drivers in the climate system (Mahmood et al. 2010; Houghton and Nassikas, 2017). Within a climate change context, there is compelling evidence of climate induced biodiversity loss (Norris et al. 2010). Due to warming and drying effects, the impacts of climate change are expected to be detrimental for not only biodiversity but also for agriculture and food systems especially in many parts of Africa (Jones and Thornton, 2009).

Thus, climate variability coupled with aggressive land-use changes hold important implications for biodiversity, and pose severe risks to livelihoods, food systems, and ecosystem health (IPCC, 2014b).

1.1. Impacts of Land-use change on woody vegetation diversity and structure

Until date, land-use changes and habitat loss (via fragmentation or a complete destruction) are the most significant ways through which humans have impacted biodiversity (Millennium Ecosystem Assessment, 2005; Pereira et al. 2012). In response to this, Protected Areas were established in West Africa as a land management tool and as a means of reducing human activities and its related disturbances (Bruner et al. 2001). Thus within this system, West African landscapes were characterized by two distinct land cover types; a natural vegetation which is usually restricted to protected areas (Naughton-Treves et al. 2005) and a communal area wherein multiple human activities and the collection of timber and non-timber forest products take place (Nacoulma et al. 2011). Unfortunately, a fast growing population, changing socio-economic conditions and rapid urbanization which is currently evident in many African countries, has created a growing pressure on even protected areas. This growing pressure has resulted in intensive modifications to how land is being used (Ouedraogo et al. 2010; Thiombiano and Kampmann, 2010). For example, the need for food and cash crops intensively transforms vast areas of forested lands into croplands leaving only a few patches of close-to-natural savanna in protected areas (Schmidt et al. 2016). Furthermore, traditional agricultural practices such as shifting cultivation which is an alternation between a short phase of cultivation and a period of fallow transforms land into a mosaic of landscapes including croplands, fallows and non-arable sites (Nacoulma et al. 2011). Patterns such as this have been shown to significantly influence vegetation dynamics in the region (Houessou et al. 2013; Ahmed et al. 2016). Furthermore, the ongoing land-use intensification pattern creates disturbances which in turn affects woody vegetation dynamics via a number of pathways including; the modification of recruitment and mortality rates (Higgins et al. 2000) or a modification in vegetation structure and plant diversity. For example; land-use change, particularly the conversion of natural vegetation to agricultural land, is associated with a decline in plant species diversity (Dumbrell et al. 2008; Tripathi et al. 2021; Bentsi-Enchill et al. 2022), the loss of tree species richness (Bobo et al. 2006; Beckmann et al. 2019), and alterations in species composition (Attua, 2003). Apart from the conversion of woodland to farmland, other land-use changes such as shortening of fallow periods and over-grazing can lead to the impoverishment of soil (Abril and Bucher, 1999). Potentially, it could also lead to an unhealthy increase in soil fertility (from the direct application of fertilizers on croplands or as direct deposits from the atmosphere) which in turn could result in (a) an increased dominance of plants adapted to nutrient-rich soils or (b) community colonization from invasive species which often out-compete other species due to their enhanced adaptive capabilities (Carey et al. 2008). Furthermore, fire which is a significant component in West African land-use systems can also negatively impact woody

vegetation diversity by either promoting the dominance of arid-tolerant species (Herrmann and Tappan, 2013) or by reducing tree regeneration (Higgins et al. 2000; 2007).

Likewise, numerous studies have investigated the impacts of land-use change on woody vegetation structure and the results have been well-documented. Some studies have reported changes in tree density, height, and basal area as a result of land use change. For example, a study by Djossa et al. (2008) in Benin found that tree density and basal area were significantly lower in agricultural lands compared to undisturbed woodlands. Furthermore, the exploitation of timber and non-timber resources particularly through the selective harvesting of woody vegetation for firewood can have immediate and pronounced effect on tree individual biomass (Luoga et al. 2002), and on the size class structure of communities (Pote et al. 2006). A recurring disturbance regime (e.g. fire or herbivory) can also limit the regeneration of trees, increase the mortality of juveniles (Mwavu and Witkowski, 2009) or reduce the density or individual biomass of trees (Luoga et al. 2002). Particularly in frequently disturbed habitats, the survival of seedling and sapling is often times dependent on their ability to persist and re-sprout (Jurisch et al. 2013). These practices can impair the vitality and regenerative abilities of trees and shrubs, or favour the expansion of more opportunistic and pioneer species. Ouédraogo et al. (2015) further described how human disturbances threaten the structure and persistence of tree and shrub populations at local and regional scales in West Africa's Sudanian savannas. It is without doubt therefore that human disturbances can play a significant role in affecting both the structure and diversity of tree populations.

1.2. Impacts of climate change on woody vegetation diversity and structure

Climate change is rapidly growing as one of the most significant threats to the world's ecosystem (Bellard et al. 2012; Scheffers et al. 2016). Studies further suggest that over the next few decades, climate change could emerge as the largest and most significant threat to biodiversity globally, surpassing habitat destruction (Leadley et al. 2010). Furthermore and in driving global patterns of woody vegetation, the impact of climate is well known. Several studies have described how latitudinal climatic gradients including precipitation, temperature and moisture influence plant establishment and distribution (e.g. McClean et al. 2005; Traoré et al. 2012; Zizka et al. 2015; Zerbo et al. 2018). It is therefore expected that drastic changes in climatic conditions will invariably affect all levels of biodiversity - from organisms to biomes such as woodlands (Parmesan, 2006). In Africa, the projected impacts of climate change will be particularly strong, with most areas expected to experience an increased frequency of extreme weather events, altered precipitation patterns, and rising temperatures (IPCC, 2014b). It is therefore expected that changes to these environmental factors will significantly drive woody plant dynamics, in particular vegetation diversity and structure. For example, future projections indicate that large proportions of suitable habitats for plant diversity will be lost in Africa as a result of climate change (McClean et al. 2005). There are also several studies that have shown how species richness will be strongly affected by climatic constraints (Field et al. 2005; Kreft and Jetz, 2007). Local studies such as the one by Mensah et al. (2016) further showed how influential, latitudinal

variation on the diversity of woody species could be. Increasing temperatures and changing precipitation patterns could also influence the diversity of woody vegetation in a number of ways; either by favouring the expansion of drought-tolerant species, or by leading to changes in tree species composition and a potential shift in dominance from forest to savanna vegetation types (Delire et al. 2008; Alexandre et al. 2011). Increased temperatures can also affect plant physiology (Jump and Peñuelas, 2005), growth rates, and the distribution of plant species (McDowell et al. 2008) which in turn can significantly influence the diversity and composition of woody vegetation. Changes in precipitation patterns can also alter the availability of water for plants (IPCC, 2014a), which in turn can facilitate plant water stress, leading to reduced growth and survival. Climate change can also intensify disturbance events particularly wildfires, an implication of which is a suppressed seedling growth (Bowman et al. 2014), highlighting the potential for altered disturbance patterns in a warmer world and how they can further shape species composition or alter ecosystem dynamics.

In a similar way, the spatio-temporal distribution of abiotic variables is a pattern that strongly influences the structure of ecosystems (Bocksberger et al. 2009). A regional study by (Assogbadjo et al. 2017) showed that climate is the major driver in determining the population structure of a woody species. Likewise, Gaitan et al. (2014) showed ways through which aridity influences vegetation structure. Changes in climatic conditions could therefore affect the structure of woody vegetation through multiple pathways including; driving the regeneration and recruitment of woody vegetation (Johnson et al. 2017), or affecting growth rates, phenology, and reproductive patterns. In addition, extreme weather events, such as floods, and storms, can cause direct damage to woody vegetation, leading to structural changes in woody vegetation. Increasing temperatures and altered precipitation patterns can also contribute to frequent and severe drought and a corresponding heat stress for woody vegetation. Research by Allen et al. (2010) provides insights into the escalating tree mortality associated with drought and heat-induced stress, thus suggesting severe implications for forest ecosystems.

1.3. Research objectives and hypothesis of the study

A crucial question thus presents itself in how these global change drivers (and their interaction) will affect the vegetation structure, and diversity of woody vegetation in West Africa's woodlands.

With regards to plant diversity and vegetation dynamics, specific trends in Africa still remain largely unknown (Heubes et al. 2013) especially in their response to land-use change (anthropogenic disturbances). The rising threat from anthropogenically-induced ecosystem disturbances (in particular, land-use intensification) has always been a serious environmental concern globally and raises strong concerns about their ecological impacts on biodiversity and other ecosystem processes. Furthermore, significant changes in the availability, composition and structure of woody vegetation would indirectly impact the sustainability of livelihoods of rural communities who derive a lot of benefits from these ecosystems. An attempt to enhance the delivery of ecosystem services while still promoting the

conservation of biodiversity therefore demands a thorough understanding of how ecological systems will be affected by disturbances (Thom and Seidl, 2016). While it is generally known that factors such as protection can play a role in conserving plant diversity (Schmidt et al. 2016) and influencing vegetation structure (Zhang et al. 2004; Wala et al. 2012), the response of ecosystem to anthropogenic disturbances is highly varied and depends on a multitude of other factors including disturbance frequency and intensity. Our study therefore aimed to explicitly show how, and to what extent taxonomic diversity and vegetation structure varies along a land-use intensification gradient (which represented different disturbance levels). Similarly, an understanding of how biodiversity and ecosystems would respond to climate change due to its strong potentials in impacting the diversity and structure of woody vegetation in West Africa's woodland. Concurrently, this is an active field of research (e.g. Gilman et al. 2010; McMahon et al. 2011) that holds significant potential for policy makers and scientists alike. Furthermore and despite significant evidence that these drivers are in a constant interaction (Oliver et al. 2014), most studies rarely consider their interactive effects on biodiversity and ecosystem services. Whilst some studies have focused on specific climatic regions (e.g. Jurisch et al. 2012; Attua and Pabi, 2013; Buba 2015; Ouédraogo et al. 2008, 2015; Balima et al. 2020), others have focused on individual components (Delire et al. 2008; Jurisch et al. 2012). To the best of our knowledge, studies on the combined effects of these global change drivers are still very limited creating an important knowledge gap. In addition, we also believe that the impact of the interactive effect of these global change factors would be higher than the sum of their individual impacts.

In this study, we applied a crossed-space-for-time substitution to understand both the isolated and interactive effects of land-use change and climate change on the diversity and structure of woody vegetation. The approach of space-for-time substitution is broadly used to exploit naturally occurring spatial gradients in understanding and possibly modelling temporal processes (Oliver and Morecroft, 2014; Blois et al. 2013). Furthermore, it is considered as one of the most efficient approaches in assessing long term disturbance effects on populations and communities (Oliver and Morecroft, 2014). By taking advantage of the steep regional gradient of climatic aridity existing in the West African region overlain by sharp local gradients of land-use intensification, climate and land-use patterns across space were used as a surrogate for temporal patterns. We combined different attributes including Species Richness, Evenness and Shannon diversity to represent an effective approach in estimating the effect of land-use change and climate change on taxonomic diversity. Likewise, in determining stand structure and the potential of regrowth within each land-use and climate type, we established demographic classifications.

1.3.1. Objectives

The main research objective that we aimed to address with this thesis is to understand the effects of land-use change and climate change on the taxonomic diversity of woody vegetation in West Africa's woodlands.

By examining the interactions between land-use change, climate change and woody vegetation dynamics, the study focused on addressing the following research questions:

- i. What are the effects of Land-use change and Climate change on Taxonomic diversity of woody vegetation; and are the effects of land-use change similar across Climate zones?
- ii. What are the effects of Land-use change and Climate change on the structure of woody vegetation; and are the effects similar across Climate zones?

1.3.2. Hypothesis

To address these questions, we formulated the following testable hypothesis;

- H1.a We expected that taxonomic diversity will successively decrease with increasing land-use intensity.
- H1.b Subsequently, we expected that with increasing climatic aridity, taxonomic diversity successively decreases.
- H1.c With regards to the interactive effect of land-use change and climate change, we expected that the negative effects of increasing climatic aridity will be more pronounced under an intensive land-use scenario i.e. we expect that the effect of climate on taxonomic diversity increases as land-use intensity increases.
- H2.a For our second hypothesis, we expected that the health of tree populations would decrease with increasing land-use intensity.
- H2.b Similarly, we expected that the negative effects of land-use intensification on the health of tree populations will be more pronounced under increasing aridity.

Scientific studies such as ours play a crucial role in alerting stakeholders to potential risks of global ecological change and can further support the development of strategies that reduce the projected impacts of both land-use change and climate change on woodland ecosystems. This is consequential because a proper quantification of how ecosystems would respond to global environmental changes is important to support the information needs for global and local policy decision-making (Amthor et al. 1998). It is perhaps more essential in the African continent where (i) human communities have very close linkages to the ecosystem, (ii) the natural ecosystems show species endemism (Thiombiano and Kampmann, 2010) and (iii) future trends suggest a continued transformation of land (Alcamo et al. 2011; Heubes et al. 2011) and worsening climate scenarios (IPCC, 2023).

2. EXPERIMENTAL DESIGN, MATERIALS AND METHODS

2.1. Study Area and Site Selection

From the Gulf of Guinea (which stretches from the Upper Guinea Region to parts of Nigeria and Cameroon) to the Southern boundary of the Sahara, the West African subregion covers a climatic gradient that is characterized by an annual precipitation varying from 250mm to 3000mm (Thiombiano and Kampmann, 2010). Similarly, land-use practices across the West African subregion are very divergent as they are often times influenced by climate, regional history and culture; thus making the study region highly suitable for simultaneously assessing land-use change and climate change effects.

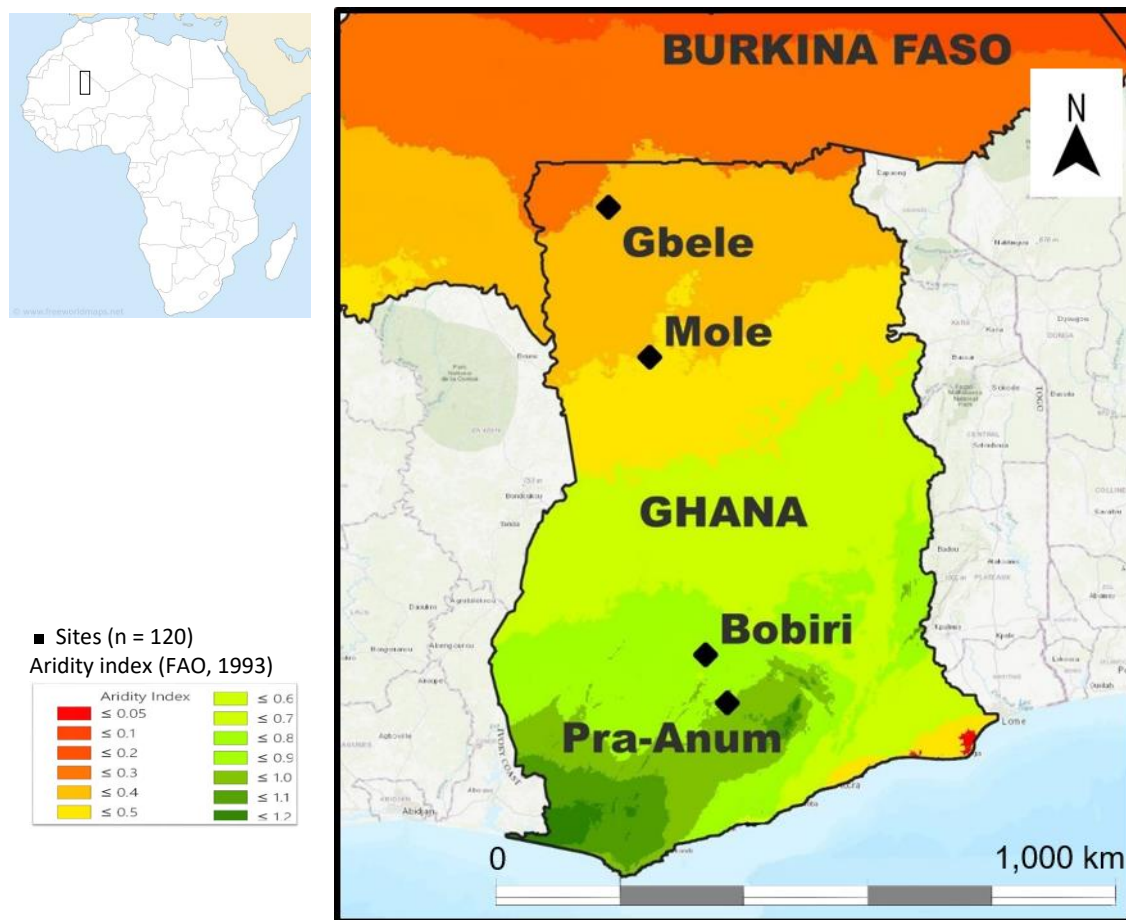


Figure 1: Study area and sampling map, showing location of Ghana in Africa (top left), and the distribution within the country of study sites. The study area was separated into two zones of increasing aridity based on FAO, 1993: A.I 0.20 – 0.50 = semi-arid, A.I 0.80 – 1.5 = Humid. In each zone, the full length of the respective land-use intensification gradient was sampled, ranging from forest to fallow to cropland.

The study, oriented along a north to south gradient was carried out in Ghana across an area covering approximately 30 ha ([Figure 1](#)). Gbele (Lat. 10°22'44"N and Long. 2°03'12"W) and Mole (Lat. 9°12'N

and Long. 2°00'W) are two of the study sites located in the North while Pra-Anum (Lat. 6°15'00"N and Long. 1°10'00"W) and Bobiri (Lat. 6°40'44"N and Long. 1°15'22"W) are the remaining two study sites which are located in the South. Climate in our study area is characterized with a unimodal rainy season in the north (from May to October), and a bimodal rainy season in the south (March to June and September to November). The mean annual precipitation (MAP) ranges from 1200mm to 1800mm in the north and south respectively (Hall and Swaine, 1981) while the annual temperature ranges from 28.2 °C in the north to 26.5°C in the south. Furthermore, our study area is characterized with the existence of a steep gradient of climatic aridity (following the classification of FAO, 1993; [Figure 1](#)), spanning an aridity index range of 0.20-0.50 (semi-arid) to an aridity index range of 0.80-1.5 (Humid). Geologically, soil types in the Northern region are the less-fertile oxisols while the relatively fertile and less-acidic ochrosols are the soil types found in the south (Brammer, 1962). Vegetation in the north consists predominantly of grassland and savanna vegetation which is characterised by the presence of an open canopy layer and a short growing season while in the south, vegetation type consists predominantly of forest lands characterised by a closed canopy, and very little to no grass layer. With regards to human activities, pastoralism, livestock rearing and subsistence farming are dominant practices in the north while activities such as commercial crop production, hunting and illegal logging are more dominant in the south.

2.1.1. Site selection

For this study, we selected three categories of land-use types namely; Forest, Fallow and Cropland. This chronology is frequently used to assess the effects of human disturbance on tree layer. A Forest is often times well protected and has a dense vegetation cover; thus it served as a reference state for disturbance states in the fallow and the cropland. It represented the lowest level of anthropogenic disturbance. The Fallow is a previously farmed Cropland that had been left unused over a period of years. The Fallow concept generally allows the land and its components including vegetation to naturally recover. For our study, the age of the Fallow used was within the range of 20 to 30 years. Fallow represented an intermediate level of anthropogenic disturbance. Land-use types that were classified as Cropland are current and actively farmed croplands. They represented the highest level of anthropogenic disturbance since they are continuously under cultivation. Selected croplands had a homogenous composition of crops such as maize and plantain as their dominant crops.

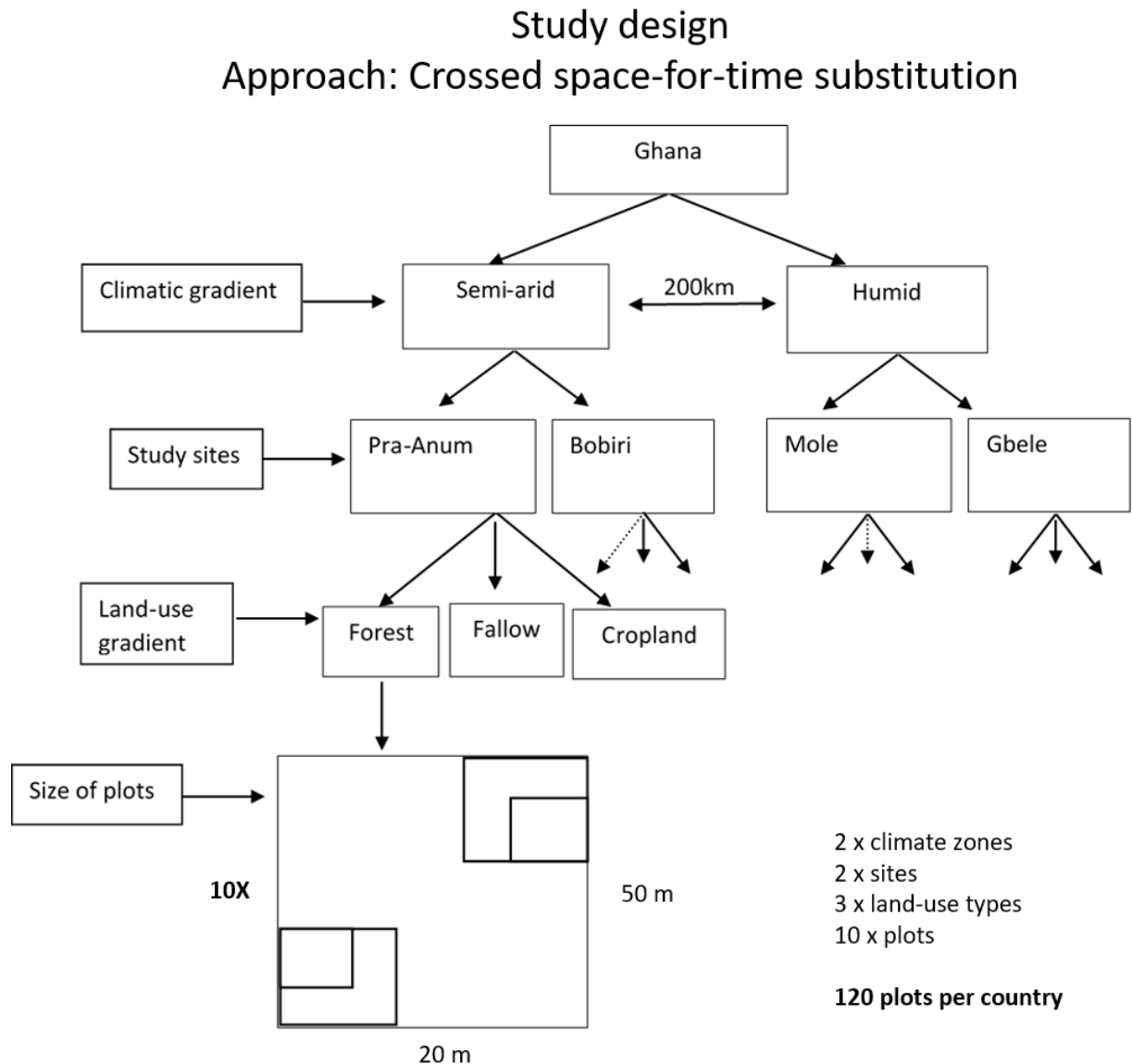


Figure 2: Conceptual framework of study design for assessing the interactive effects of land-use change and climate change on the diversity and structure of woody vegetation in Ghana.

2.2. Sampling design and data collection

With our sampling design, we intended to capture land-use gradients along the regional climatic gradient. For this reason, we stratified our sampling into two zones of climatic aridity (Figure 1). Within each climate zone, we compared the varying levels of land-use intensification that can be observed in West African landscapes namely; Forest, Fallow and Cropland (Figure 4). Vegetation surveys and data collection were conducted between March and June 2022. Ten independent observation plots were established per site for each land-use type (30 plots in total for each site). Across all sites, a total of 120 independent observation plots were established (4 sites along the climate gradient x 3 land-use gradient

x 10 replicates; [Figure 2](#)). In Forest, each plot had a size of 20 m x 50 m, while in the Fallow and Croplands, we established plots with a size of 50 m x 50 m. Differences in plot sizes between the land-use types were due to the density of trees present, as we observed that Cropland and some Fallow had a low tree density. Within the plots, we established a bigger subplot with a size of 10 x 10 m, and a smaller subplot of 5 x 5 m was nested within the 10m x 10m subplot at two diagonals of the 20 x 50 m / 50 x 50 m ([Figure 3](#)). For the purpose of measuring different vegetation components, we employed a nested plot design in our study [[Figure 3](#)]. To avoid pseudo replications during our data collection, we observed a minimum distance of 200m between each established plot. We also ensured that there was a considerable distance between each sampling plot and already established roads/transect in order to avoid likely edge effects. All plot establishment procedures was done with the aid of a measuring tape, ranging poles, and the application of Pythagoras' theorem.

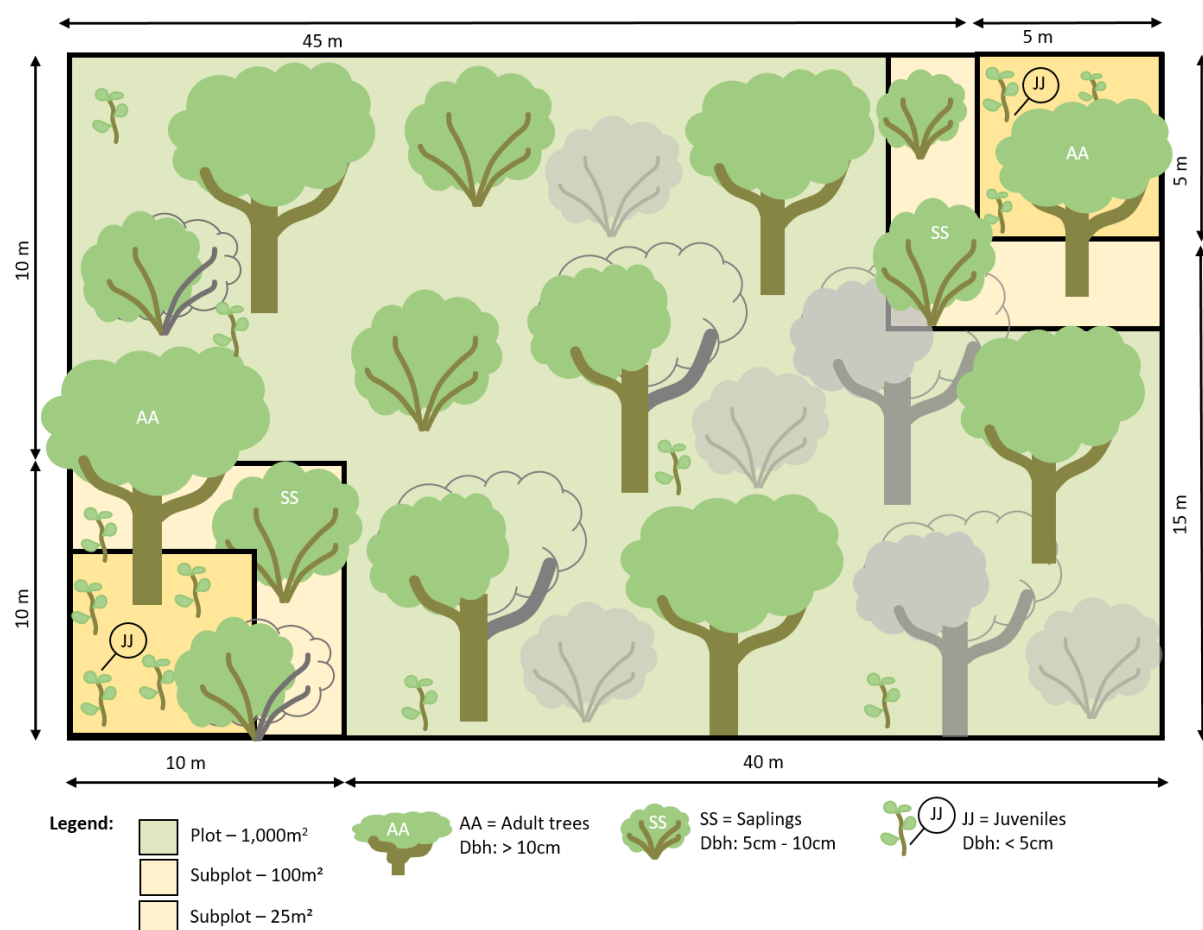


Figure 3: sampling design showing the plot layout and sampling strategy for the tree inventory. In the 1000 m² plot, all individuals with a dbh >10cm are recorded; in the 100 m² subplot, all individuals within a dbh range of 5cm-10cm are recorded; in the 25 m² subplot, all individuals with a dbh <5cm are recorded. Except for croplands which had a plot size of 2500m² (50 x 50m), all other land-use types had a plot size of 50 x 20 m.

2.2.1. Vegetation sampling

From all plots, we classified (based on diameter classes) and recorded woody tree species across three ecologically defined age classes (Adult, Saplings and Juveniles) [Table 1]. Woody, and self-supporting single-stemmed plants with a diameter at breast height (hereafter referred to as dbh) above 10 cm were classified as Adults. These were sampled on the main plot (plot size of 20 x 50m OR 50 x 50m), identified at the species level and their dbh and height measurements were recorded. Woody and self-supporting multi-stemmed or single stemmed plants with a dbh range of 5 cm - 10 cm were classified as Saplings. Likewise, these were identified at the species level and their dbh and height measurements were recorded. However, these were only sampled in the subplot (plot size of 10 x 10 m). Self-supporting multi-stemmed or single stemmed plants with a dbh range lesser than 5 cm were classified as Juveniles. They were also identified at the species level but only their height measurements were recorded. Juveniles were identified and sampled on the smaller subplots (plot size of 5 x 5m).

Table 1: Growth classes of woody tree species separated into sampling effort based on ecologically defined thresholds of stem diameter.

Growth class	diameter at breast height (dbh)	Measured on (sampling level)
Adults	> 10cm	50 x 50 m or 50 x 20 m
Saplings	5 - 10cm	10 x 10 m
Juveniles	< 5cm	5 x 5 m

2.3. Data processing and Analysis

2.3.1. Objective 1: Effect (isolated and interactive) of land-use change and climate change on Taxonomic diversity

For a better understanding of, and for an accurate measurement of the biological diversity found within the different land-use types and climatic zones, we computed a variety of taxonomic diversity indices such as Species Richness, Pielous's Evenness and Shannon-Wiener's Diversity Index (H') – these are some of the most basic proxies used in quantifying alpha diversity (Magurran, 2004; Fedor and Zvaríková, 2019). We calculated taxonomic diversity, specifically Shannon-Wiener's Diversity Index (H') based on the relative abundance and the relative frequency of each species. In calculating the relative abundance, allometric equations were used to estimate individuals' aboveground biomass from recorded biometric data. We used the widely adopted allometric equation model from Chave et al. (2014) to estimate the biomass of adult individuals and saplings based on their recorded height and dbh values. For estimating the aboveground biomass with the allometric model of Chave et al. (2014), we also retrieved species' specific wood density (SWD) values using the BIOMASS package (Réjou-

Méchain et al. 2017) in the opensource programs R and RStudio. Subsequently and for juveniles, we used the allometric equation from Mbow et al. (2013) to estimate biomass from tree height (h).

2.3.2. Objective 2: Effect (isolated and interactive) of land-use change and climate change on structural composition and health of tree populations

Size class distributions (SCD) are a widely used tool especially when describing the effect of land use on population structures (Lykke, 1998a; Jurisch et al. 2012; Ouédraogo et al. 2015). They are also useful in predicting growth trends which can help facilitate better decision making and sustainable forest management. One popular assumption with size class distributions are that a negative SCD slope indicates the occurrence of a higher number of individuals in the lower stem diameter classes (Juveniles, Saplings); an interpretation of which would be that the population is quite stable with sufficient recruitment potential and vice versa (Condit et al. 1998; Lykke, 1998a; Jurisch et al. 2012; Ouédraogo et al. 2015). We grouped individuals of species into equally distant stem diameter classes of 5 cm width following similar studies by Condit et al. (1998) for tropical forest regions and Bognounou et al. 2009, Jurisch et al. 2012, Ouédraogo et al. 2015 for savanna forest regions. All individuals with a dbh larger than 60 cm were grouped into a single class (≥ 60 cm). We therefore established the size class distributions (SCD) of our measured woody vegetation based on these mentioned studies. In consideration of the fact that we had set up sampling plots of various sizes for each age class, we applied an upscaling factor (see [Formulas](#) in [Appendix](#)) to be able to express each stem diameter class on a per unit area basis (in individuals ha^{-1}). The number of individuals in each size-class (N_i) and the size-class medians were used in a log-linear regression as the dependent and independent variables. Because some size-classes had no individuals, an $\ln(N_i + 1)$ transformation was carried out as suggested by Lykke (1998a). The slopes of the size-class distribution (SCD) for different land-use types were determined using approaches already described by Condit et al. (1998) and Lykke (1998a). These SCD slopes and their coefficients of determination (R^2) were thereafter utilized as indicators of population structure. Negative slopes with SCDs skewed toward the lower stem diameter classes indicates that the population was stable and there was sufficient recruitment. In contrast to this, flatter-shaped slopes with SCDs that are bell-shaped would indicate either a relatively high number of adult trees or that there was little recruitment. Positive slopes and SCDs skewed towards the higher stem diameter classes indicate ageing populations or that there was low recruitments. To allow for visual comparisons, size-class distributions (SCDs) were constructed and displayed graphically.

2.3.3. Data analysis

All data collected at the plot-level was aggregated and the mean values were computed for each land-use type and climate zone. All statistical and data related analysis were conducted using the opensource programs R (R Core Team, 2022) and RStudio (R Studio Team, 2023) with statistical packages dplyr (Wickham et al. 2023), vegan (Oksanen et al. 2022), BiodiversityR (Kindt and Coe, 2005), BIOMASS (Réjou-Méchain et al. 2017). We used the packages ggplot2 (Wickham, 2016), and ggpubr (Kassambara,

2023) for the data visualization and in producing graphs. We tested for the significance of land-use type and climate effects on woody vegetation structure and diversity using a Two-way ANOVA, followed by Tukey HSD post hoc tests for multiple or pairwise comparisons. All data were checked for normality distribution and log-transformed where necessary to meet assumptions of homoscedasticity. We visualized differences with boxplots and histograms.



Figure 4. Photographic examples for the prominent land-use types that were observed in the study area. A) Forest; B) Fallow; C) Cropland.

3. RESULTS

Across all land-use types and from an approximate area of 28ha, we recorded a total number of 9133 woody plants; 3197 adult individuals (1897 in the humid climatic zone and 1300 in the semi-arid climatic zone respectively), 444 saplings (272 in the humid climatic zone and 172 in the semi-arid climatic zone) and 5492 juveniles (4159 in the humid climatic zone and 1333 in the semi-arid climatic zone).

Table 2: Summary of the taxonomic diversity for woody vegetation in different land-use types across a climatic gradient of humid to semi-arid in Ghana. Mean values were calculated based on values derived from 20 individual plots within each land-use type and across both climate zones

Climate zone	Aridity index	Land_use_type	mean.Richness	mean.Evenness	mean.Shannon
Humid	0.80 – 1.5	Forest	41.2	0.7	2.4
Humid	0.80 – 1.5	Fallow	19.8	0.6	1.3
Humid	0.80 – 1.5	Cropland	4.5	0.7	0.9
Semi-arid	0.20 – 0.50	Forest	12.1	0.6	1.6
Semi-arid	0.20 – 0.50	Fallow	11.8	0.6	1.5
Semi-arid	0.20 – 0.50	Cropland	2.0	0.6	0.2

3.1. Effect of Land-use change on Taxonomic diversity

3.1.1. Species Richness

Land-use change had a significant effect on species richness and this was similar across the two climate zones ($F = 91.07$, $P < 0.0001$; [Table 3](#)). This effect was particularly evident in the humid zone, as the richness value of each land-use type significantly varied from the other [Fallow - Forest ($P < 0.0001$), Fallow - Cropland ($P < 0.0001$), Cropland - Forest ($P < 0.0001$)]. Furthermore, species richness in the humid zone successively decreased along a land-use intensification gradient by 89 % when land is converted from Forest to Cropland and makes a complete recovery (by 340%) when these croplands are abandoned and become Fallow ([Figure 5](#)).

A similar pattern was recorded in the semi-arid zone, with species richness decreasing by 83 % when land is converted from Forest to Cropland and subsequently increasing by almost 500% when croplands are abandoned to Fallow ([Figure 5](#)). The mean value for species richness in Forest plots was 41 in the humid zone and 12 in the Semi-arid zone while in Fallow plots, the mean value for species richness was 20 the humid zone and 12 in the Semi-arid zone. Cropland plots from both climate zones had the lowest species richness with a mean value of 5 in the humid zone and a mean value of 2 in the semi-arid

zone. Forest in the humid zone had the highest mean value for species richness, whereas the lowest mean value for species richness was recorded from Cropland in the semi-arid zone ([Table 2](#)).

3.1.2. Species Evenness

Land-use change had no considerable effect on the distribution of species across both climate zones as species evenness in the various land-use types did not significantly vary ($F = 1.10$, $P = 0.339$; [Table 3](#)). Evenness values were high and similar within the different land-use types with a global mean value of 0.6 ([Table 2](#)).

3.1.3. Shannon-Weiner Diversity

The diversity index (Shannon's Index H') varied considerably across land-use types ($F = 68.13$, $P < 0.0001$; [Table 3](#)) along the different climate zones. In the humid climate zone, the diversity of species significantly differed between Forest and the remaining land-use types i.e. Fallow and Cropland [Forest - Fallow ($P < 0.0001$), Forest - Cropland ($P < 0.0001$)]. When Forest was converted to other use types such as Cropland, the diversity of species decreases by 63%. In contrast, when previously farmed croplands are abandoned and left to Fallow, the diversity of species recovers but by only 44%. Statistically, the difference in the Shannon diversity values for Cropland and Fallow was not significant ($P > 0.05$) and as a result, the recovery of species diversity in the humid climate zone from Fallow was considered to be relatively low. Furthermore, when these land-use types are compared to Forest, Forest still had the highest Shannon diversity value of 2.4.

In the semi-arid zone, Land-use intensification had a higher impact on species diversity as we observed a decrease in diversity values by 88% when Forest was converted to Cropland ([Figure 5](#)). Similar to the humid zone, species diversity in the semi-arid zone had a high recovery rate of about 650% when previously farmed croplands were abandoned and left to Fallow. We therefore considered the recovery of species diversity in Fallow in the semi-arid zone to be complete because when compared to Forest ($H' = 1.6$), the Shannon diversity values in Fallow ($H' = 1.5$) showed no significant difference [Forest - Fallow ($P > 0.05$)]. In general, the most diverse land-use type was Forest in the humid zone, whereas the least diverse land-use type was Cropland in the semi-arid zone ([Table 2](#)).

Land-use effects on Taxonomic diversity

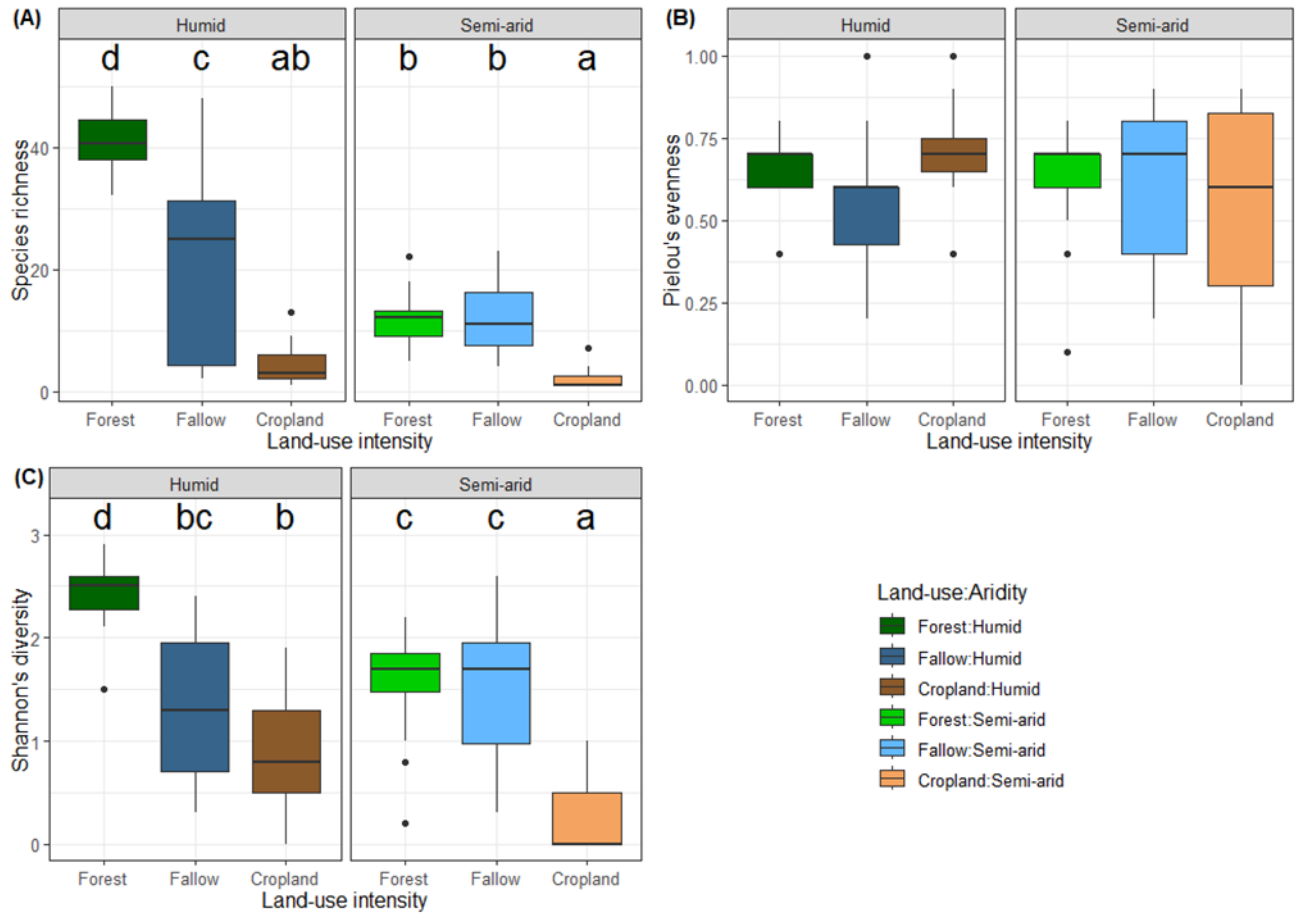


Figure 5: A box plot comparison of the taxonomic diversity of woody vegetation across a land-use intensification (Forest – Fallow – Cropland) gradient. (A) Species Richness (B) Pielou's Evenness (C) Shannon-Weiner Diversity Index. Species richness and Shannon-Weiner diversity decreases along a land-use intensification gradient across both climate zones. The line within each box represents the median value. Superscript letters indicate significant differences according to Tukey's HSD post hoc test while same letters indicate that there is no significant difference. Note different Y axis scales

3.2. Effect of Climate change on Taxonomic diversity

3.2.1. Species Richness

Climate change proved to have an effect on the number of different species found in each land-use type as Species richness considerably increased from North to South along a Climatic gradient ($F = 101.60$, $P < 0.0001$; Table 3). Species richness decreased in Forest by 71% ($P < 0.0001$), in Fallow by 40% ($P < 0.05$), and in Cropland by 56% ($P > 0.05$) respectively, as climatic conditions become less humid and more arid (Figure 6). On average, a higher species richness was recorded across land-use types in the humid climate zones as compared to land-use types in the semi-arid climate zone (Table 2).

3.2.2. *Species Evenness*

Similar to land-use change effect, climate change showed no significant effect on the distribution of species abundance in each land-use type ($F = 0.12$, $P > 0.05$; [Table 3](#)). Land-use types in the humid and semi-arid zones had a global average value of 0.6 for species evenness ([Table 2](#)).

3.2.3. *Shannon-Weiner Diversity*

The analysis revealed an effect of climate change on species diversity as Shannon's diversity values differed considerably between the two climate zones with a North to South increase in species diversity ($F = 16.40$, $P < 0.0001$; [Table 3](#)). An in-depth comparison of the effect of climate change across land-use types revealed that while there was an observable decrease of species diversity in Forest and Cropland (by 33%; $P < 0.0001$ and 78%; $P < 0.05$ respectively) with increasing aridity, a contrasting pattern is observed in Fallow as there is instead an increase in species diversity as climatic conditions became more arid (by 15%, $P > 0.05$; [Figure 6](#)).

3.3. **Interactive effect of Climate and Land-use change on taxonomic diversity**

3.3.1. *Species Richness*

The results of our analysis showed that the interaction of climate and land-use change had a significant effect on the number of different species present ($F = 33.40$, $P < 0.0001$; [Table 3](#)). It was observed that the negative effects of increasing aridity are more pronounced under intensive land-use scenario as Cropland in the semi-arid zone had the lowest species richness (average of 2 species per plot) and differed significantly from Fallow in the semi-arid climate zone [Cropland: Semi-arid - Fallow: Semi-arid ($P < 0.001$), [Figure 6](#)]. Our results also showed a reduction in the number of species by 95% when there is a transition from less intensively used land in humid climate zones to a high land-use intensification in semi-arid climate zones ([Figure 6](#)). With regards to recovery, while Cropland in the semi-arid zone show a recovery of almost 500% when they are left to fallow, a higher recovery of approximately 900% is expected if the climatic conditions of these same land-use types improve to more humid conditions.

3.3.2. *Species Evenness*

Our results showed that the interaction of climate and land-use change had no significant effect on the distribution of species abundance ($F = 1.84$, $P > 0.05$; [Table 3](#)).

3.3.3. *Shannon-Weiner Diversity*

Similar to species richness, the results of our analysis showed an interactive effect of climate change and land-use change on species diversity ($F = 8.84$, $P < 0.0001$; [Table 3](#)). As observed, the negative effects of increasing climatic conditions were also more pronounced under an intensively used land as

Cropland in the semi-arid zone had the lowest species diversity (mean H' of 0.2) and differed significantly from Fallow in the semi-arid [Cropland: Semi-arid - Fallow: Semi-arid ($P < 0.001$), [Figure 6](#)]. Our results also show that there is a reduction in species diversity by 92% when there is a transition from a less intensively used land in humid climate zones to a scenario of high land-use intensification and increasing climatic aridity (From Forest: Humid to Cropland: Semi-arid, [Figure 6](#)). With regards to recovery (Cropland to Fallow), we observed a slightly different pattern as improving climatic conditions did not lead to a higher recovery of diversity (Cropland: Semi-arid to Fallow: Semi-arid; recovery of 650%, Cropland: Semi-arid to Fallow: Humid; recovery of 550%).

Table 3: Summary of the analysis of variance comparison on the effect of Land-use, Climate, and Land-use*Climate (predictor variables) on Species richness, Species Evenness, and Shannon-Weiner Diversity (response variables). Their significance levels are also given.

Estimation parameters		df	Sum Sq	Mean Sq	statistic	p.value	sign
Species richness	Land_use_type	2	9883	4941	91.07	1.39e-23	***
	Aridity_class	1	5513	5513	101.60	4.47e-17	***
	Land_use_type:Aridity_class	2	3624	1812	33.40	6.27e-12	***
	Residuals	104	5643	54			
Species evenness	Land_use_type	2	0.081	0.04031	1.095	0.339	ns
	Aridity_class	1	0.004	0.00435	0.118	0.732	ns
	Land_use_type:Aridity_class	2	0.136	0.06789	1.844	0.164	ns
	Residuals	91	3.349	0.03681			
Shannon weiner's diversity	Land_use_type	2	40.18	20.090	68.132	1.23e-19	***
	Aridity_class	1	4.84	4.837	16.403	9.88e-05	***
	Land_use_type:Aridity_class	2	5.21	2.607	8.841	0.000284	***
	Residuals	104	30.67	0.295			

Significance codes: 0 '***', 0.001 '**', 0.01 '*', >0.05 'ns'. [ns= not significant]

Interactive effects of Land-use and Climate on Taxonomic diversity

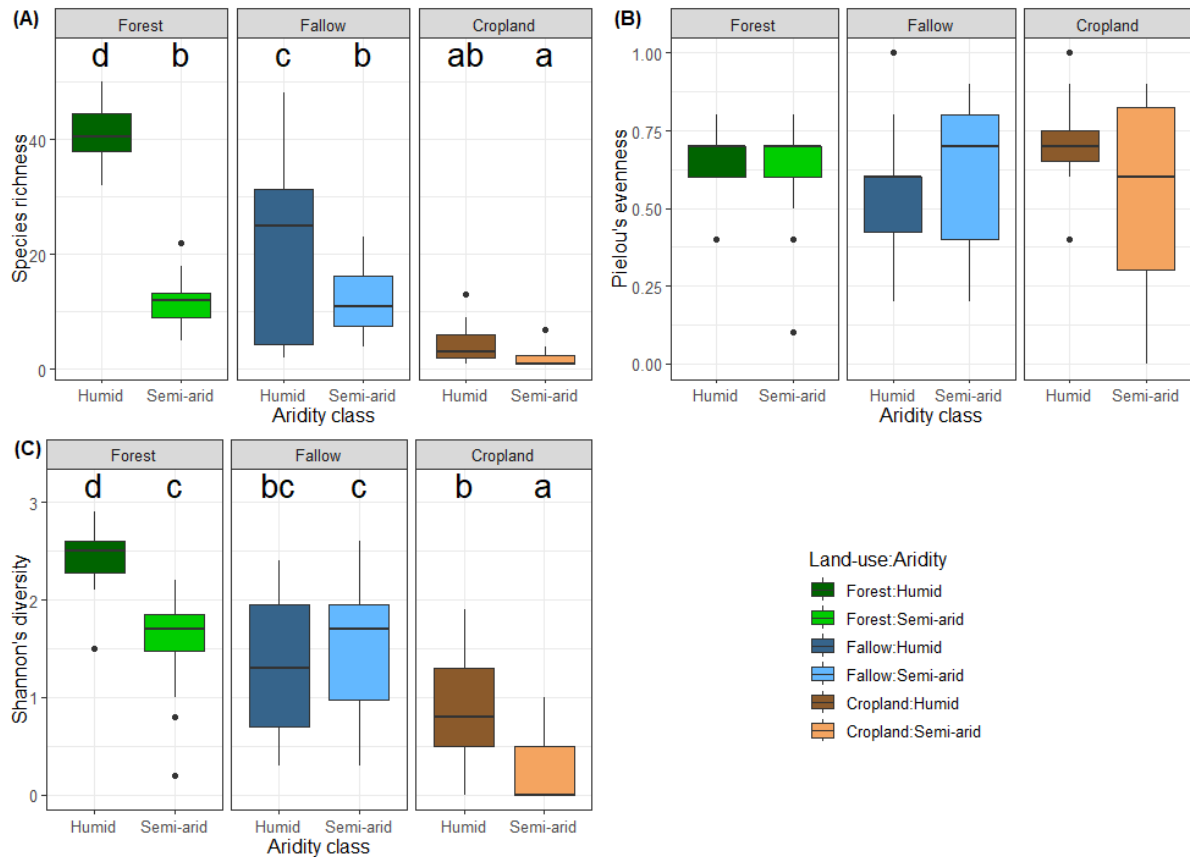


Figure 6: A box plot comparison of the taxonomic diversity of woody vegetation recorded across a climatic (Humid – Semi-arid) and land-use intensification (Forest – Fallow – Cropland) gradient in Ghana. (A) Species Richness (B) Pielou's Evenness (C) Shannon-Weiner Diversity Index. The line within each box represents the median value. Superscript letters indicate significant differences according to Tukey's post hoc test while same letters indicate that there is no significant difference. Note different Y axis scales.

3.4. Effect of Land-use change on the health of Tree Populations

Land-use change had a strong influence on the health of tree populations as vegetation structure differed considerably across all land-use types.

3.4.1. Population density per land-use type

The results of our analysis showed that the density of woody vegetation varied within the different land-use types across both climate zones (Table 4). Forest within the humid zone had the highest density of woody vegetation (968 individuals per hectare) with an estimated population distribution of 633 adults, 117 saplings, and 218 juveniles while in Fallow, the density of woody vegetation was 492 individuals per hectare with an estimated distribution of 271 adults, 85 saplings and 136 juveniles. Cropland within the humid climate zone however had the lowest density of woody vegetation (135 individuals per hectare) with an estimated distribution of 120 adults, 15 saplings, and zero juveniles. In the semi-arid

zone, the population density of Forest was estimated at 644 individuals per hectare with a distribution of 339 adults, 116 saplings, and 189 juveniles while the population density of Fallow and Cropland was estimated at 530 and 191 individuals per hectare respectively (distribution of 266 adults, 78 saplings and 186 juveniles for Fallow and a distribution of 153 adults, 7 saplings and 31 juveniles for Cropland).

Fallow within the semi-arid zone however, had the highest percentage of juvenile to adult proportions (49.8%) followed by Forest within the same climate zone (47.4%) while Cropland within the humid zone had the lowest percentage of juvenile to adult proportions (11.1%). The percentage of juvenile to adult proportions for the remaining land-use types are 44.9% (Fallow: Humid), 34.6% (Forest: Humid), and 19.9% (Cropland: Semi-arid) respectively ([Table 4](#)).

3.4.2. Size-class distributions

The shape of the diameter class distribution varied considerably between land-use types and across the climate gradient (see [Appendix Figure 9](#)) indicating that land-use change strongly influenced size-class distributions. In the humid zone, visual observations of the vegetation structure in Forest showed a reverse “J” shaped curve (see [Appendix Figure 9](#)). Further analysis of the diameter-size class distribution showed that Forest had a significantly negative SCD slope ($Slope = -2.24$, $R^2 = 0.67$, $P = <0.0001$; [Table 5](#)) indicating that there were more individuals in the smaller size classes (dbh <10cm) relative to the larger size classes (dbh >10cm). A similar pattern (inverse J-shaped curve) was also observable for Fallow in the humid zone. The diameter-size class distribution here showed a structure skewed towards the lower classes and a significantly negative slope ($Slope = -1.90$, $R^2 = 0.45$, $P = <0.0001$) indicating a stable population and the occurrence of sufficient recruitment. A slightly contrasting pattern was however observed in Cropland from the humid zone as visual observation showed a bell-shaped curve with close to equal numbers of plants in both small and large size-classes (see [Appendix Figure 9](#)), indicating little to low recruitment. Furthermore, even though the slope of the SCD was negative, it was not significant and was almost flat ($Slope = -0.04$, $R^2 = -$, $P = 0.5$). In the semi-arid zone, the vegetation structure of Forest was similar to that of its counterpart in the humid zone. The size class distribution in this land-use type was skewed towards the lower classes and it also had a significantly negative slope ($Slope = -2.50$, $R^2 = 0.72$, $P = <0.0001$). Likewise, Fallow had the same pattern with a significantly negative size class distribution slope ($Slope = -2.30$, $R^2 = 0.69$, $P = <0.0001$) and a distribution skewed towards the lower diameter classes. In Cropland, SCD slopes were somewhat flat but significant ($Slope = -0.41$, $R^2 = 0.07$, $P = <0.0001$) and visual observation showed a bell-shaped size-class distribution with similarly close to equal numbers of plants in both small and large size-classes.

Paired t-tests comparing the size class distribution slopes of the different land-use types showed that they differed significantly within the land-use types and across the climate zones ($F = 124.27$, $P = <0.0001$; [Table 6](#)). The respective differences (see [Figure 7](#)) were observed across both climate zones in: Forest and Cropland ($P <0.0001$), in Fallow and Cropland ($P <0.0001$), but not between Forest and Fallow ($P >0.05$).

3.5. Interactive effect of Climate and Land-use on the health of Tree Populations

The results of our analysis showed no interactive effect of climate and land-use change on the health of tree populations ($F = 0.142$, $P > 0.05$; [Table 6](#)). Paired t-tests comparing the different size class distribution slopes showed that the negative effects of land-use intensification on tree population health are **not** more pronounced with increasing climatic conditions ([Figure 8](#)).

Table 4: Structural attributes of the different land-use types across a climatic gradient of humid to semi-arid in Ghana. Woody vegetation density per hectare (normalised using a $\log_{10}(x + 1)$ transformation) and juvenile percentages are shown for the different land-use types.

	Climate									
	Humid (A.I: 0.80 – 1.5)					Semi-arid (A.I: 0.20 – 0.50)				
	Woody vegetation density									
	Adults	Saplings	Juveniles	% Juv.	Density (N.ha ⁻¹)	Adults	Saplings	Juveniles	% Juv.	Density (N.ha ⁻¹)
									.	
Land-use										
Types										
Forest	633	117	218	34.6	968	339	116	189	47.4	644
Fallow	271	85	136	44.9	492	266	78	186	49.8	530
Cropland	120	15	0	11.1	135	153	7	31	19.9	191

Table 5: Size-class distribution (SCD) analyses for the different land-use types across a climatic gradient of humid to semi-arid in Ghana. Regression slopes of each land-use type were calculated by plotting the size-class midpoints against the number of individuals found in each size-class.

	Climate							
	Humid (A.I: 0.80 – 1.5)				Semi-arid (A.I: 0.20 – 0.50)			
	Slope	Intercept	R ²	P	Slope	Intercept	R ²	P
Land-use Types								
Forest	–2.24	11.14	0.67	<2.2e-16	–2.50	10.77	0.72	<2.2e-16
Fallow	–1.90	8.20	0.45	<2.2e-16	–2.30	9.68	0.69	<2.2e-16
Cropland	–0.04	1.00	–	0.5	–0.41	2.10	0.07	9.13e-06

Table 6: Summary of the analysis of variance comparison on the effect of Land-use, and Climate*Land-use (predictor variables) on size class distribution slope (response variable). Their significance levels are also given.

Estimation parameter	df	Sum Sq	Mean Sq	statistic	p.value	sign
SCD slope						
Land_use_type	2	90.93	45.47	124.273	<2.2e-16	***
Aridity_class	1	3.13	3.13	8.546	0.00425	**
Land_use_type:Aridity _class	2	0.10	0.05	0.142	0.86811	
Residuals	104	38.05	0.37			

Land-use effects on the health of Tree Populations

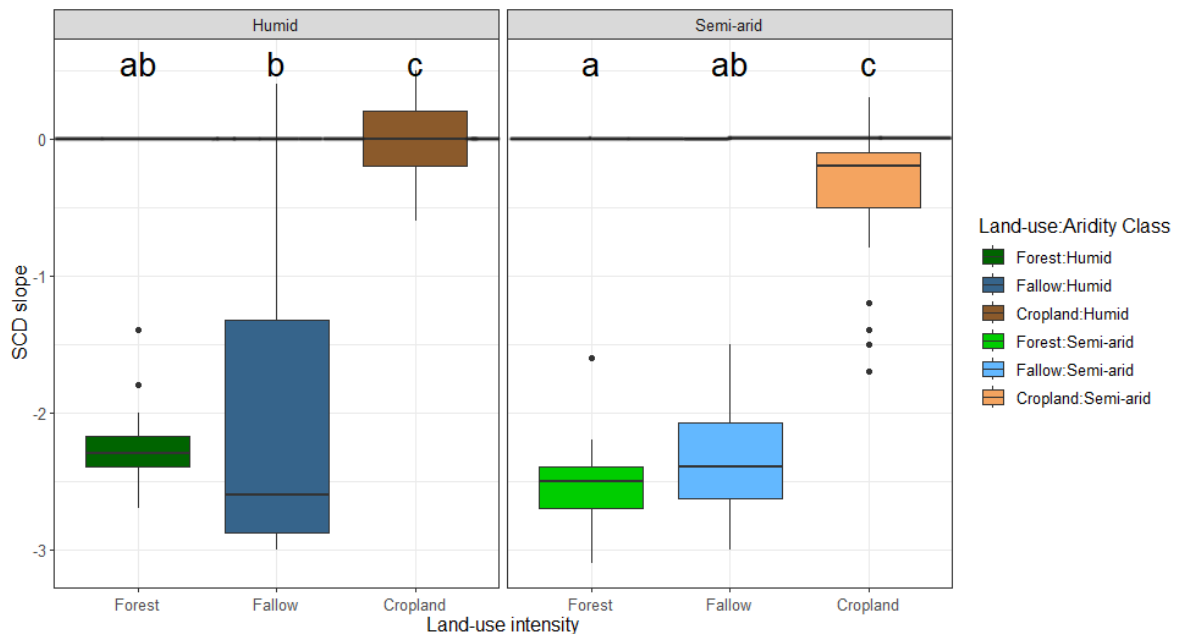


Figure 7: A box plot comparison of the size class distribution slopes across a land-use intensification (Forest – Fallow – Cropland) gradient in Ghana. The grey horizontal line in the box represents the null value. Flat slope values directly on the line indicate little recruitment, positive values above the line imply ageing population and low recruitment, while negative values below the line imply a stable populations and sufficient recruitment. Superscript letters indicate significant differences according to Tukey's post hoc test while same letters indicate that there is no significant difference.

Interactive effects of Climate and Land-use on the health of Tree Populations

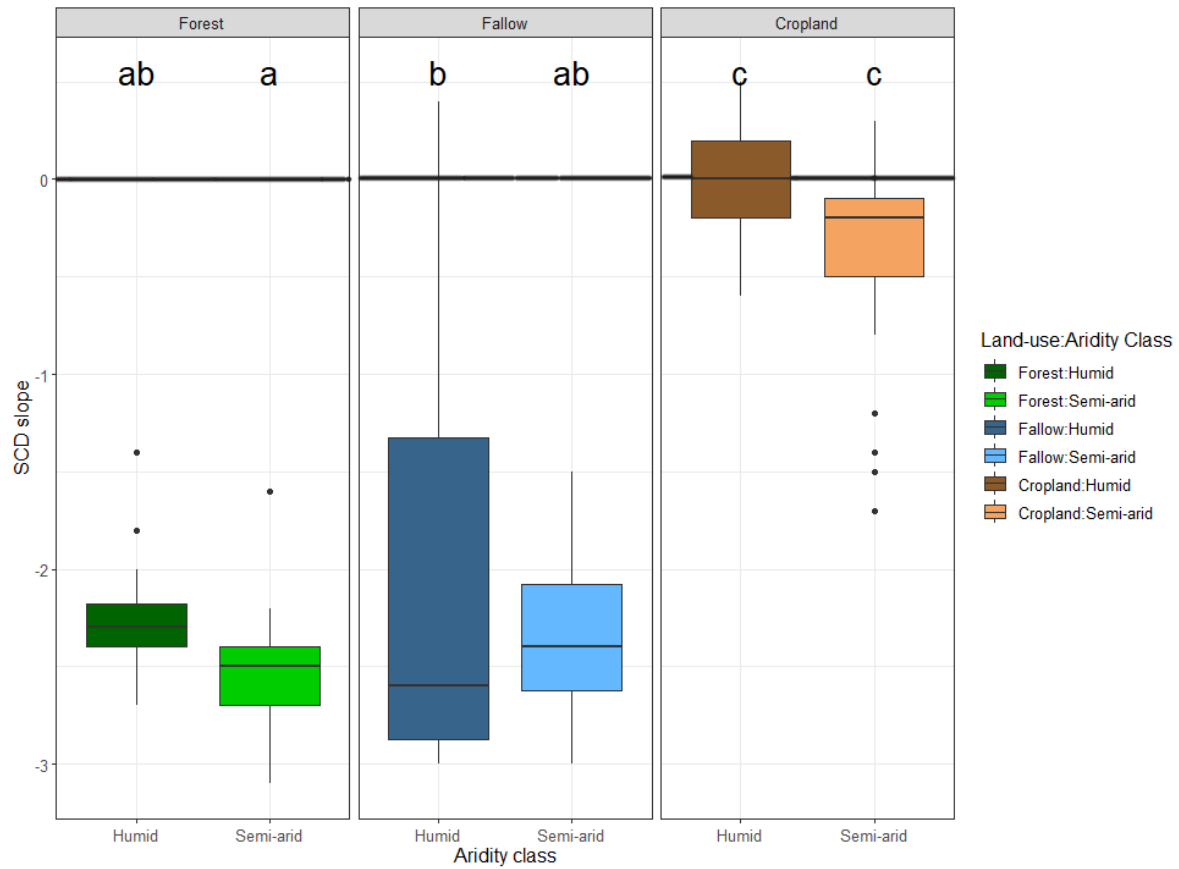


Figure 8: A box plot comparison of the size class distribution slopes showing the interactive effects of climate and land-use on the health of tree populations in Ghana. The grey horizontal line in the box represents the null value. Flat slope values directly on the line indicate little recruitment, positive values above the line imply ageing population and low recruitment, while negative values below the line imply a stable populations and sufficient recruitment. Superscript letters indicate significant differences according to Tukey's post hoc test while same letters indicate that there is no significant difference.

4. DISCUSSION

The main objectives of this study was to understand the effects of land-use change and climate change on the diversity and structure of woody vegetation in West Africa's woodlands. Specifically, we aimed to investigate if: (i) land-use intensification leads to a reduction in taxonomic diversity and whether the effects of land-use change are similar across climate zones, (ii) land-use intensification affects the structure of tree populations and whether these effects are similar across climate zones.

4.1. H1a: Land-use effect on Taxonomic diversity

Our first expectation was that taxonomic diversity would progressively decrease with an increase in land-use intensity across both climate zones. This was confirmed as the results of our study revealed a clear and significant decrease in taxonomic diversity along a land-use intensification gradient (measures of biodiversity in the focus of our discussion are Species richness and Shannon-Weiner diversity). Croplands in both climate zones had the lowest number of species and woody vegetation diversity indicating a high sensitivity of these land-use types to the impact of anthropogenic disturbances. The overall diversity of woody vegetation was highest in the land-use category of Forest within the humid zone and we suggest that this is as a consequence of the high species richness simultaneously recorded in this land-use type thus signalling a possible correlation between diversity and species richness. Furthermore, the varying responses of the land-use types to taxonomic diversity show the degree to which anthropogenic disturbances can influence woody vegetation diversity. Our findings are consistent with several studies within the region and around the world which also demonstrated a reduction in the taxonomic diversity especially for agricultural lands. Within the region, a recent study by Balima et al. (2020) highlighted the negative effect of agricultural land-use related disturbances; these land-use types had the lowest species richness and woody vegetation diversity in comparison to less intensively used and protected land-use types while Ky-Dembele et al. (2019) reported that croplands and agricultural areas in Mali had lower species richness compared to natural habitats. In Ghana, Addo-Fordjour et al. (2009) and Bentsi-Enchill et al. (2022) showed that plant species diversity quantitatively reduced along a disturbance gradient with the number of species higher in undisturbed plots. Outside the region, a study by Billeter et al. (2008) found that as agricultural land-use intensified in a temperate forest ecosystem, there was a substantial decline in species richness and other diversity indices. Similarly, Villa et al. (2018) indicated a marked reduction in biodiversity as deforestation increased and natural habitats in the Amazon rainforest were converted into croplands. All of these studies parallel our results, emphasizing that taxonomic diversity decreases along a disturbance gradient.

We suggest that the reason why the Forest land-use type had the highest taxonomic diversity compared to other land-use types was because all Forest plots were established in protected areas which in general are less targeted by humans and consequently have very little human activity. Likewise, we assume that the reason why croplands had a reduced capacity to support diverse woody vegetation is as a result of

their non-protected status. Oftentimes, the transition or conversion of land into croplands is characterized with the destruction and fragmentation of habitats, a practice that is known to reduce biodiversity (Sambaré et al. 2011; Dudley et al. 2017). Consequently, species that once depended on the original ecosystem may be lost as croplands replace natural habitats, leading to a decline in species richness. In addition to this, the practice of slash-and-burn agriculture in West Africa, whereby large forested areas are cleared for the cultivation of crops not only eliminates forest cover directly but also contributes to soil degradation and nutrient loss (Burton et al. 1989) which can further hinder the ability of woody vegetation to thrive. Additionally, the simplified structure of croplands, which is often a combination of fewer plant species and a lack of mature trees and shrubs, can contribute to a decrease in the diversity of woody vegetation (Vandermeer, 2009).

4.1.1. Land abandonment as a reservoir for biodiversity

From the result of our investigation on land-use effects on woody vegetation diversity, two particular trends from the semi-arid zone stand out. The first is that species richness makes a significant recovery when croplands are abandoned (in our study scenario, for a minimum period of 20 years) and secondly, that forests and fallows do not differ significantly in the diversity of woody vegetation. The observed improvement in species richness and diversity of abandoned croplands (particularly those in the semi-arid zone) suggest that diversity can recover if degraded lands are left to fallow. Historically, fallows have been a major component of traditional land-use systems in the tropics (Ruthenberg, 1980) and are highly valued due to their potential in restoring soil fertility (Szott et al. 1999) or even acting as a reservoir for non-timber forest products which often times are used to generate income off-farm and off-season (Hirai, 2014). Our finding is consistent with the finding of Gonçalves et al. (2017) and Daskalova and Kamp (2023) which both describe the potentials of biodiversity recovery after agricultural abandonment. In West Africa, studies by Augusseau et al. (2006) and Anglaaere et al. (2011) have also showed that these land-use types have a species diversity even greater than that of natural forests, thus reemphasizing the high potential of fallows as reservoirs of biodiversity. In addition to those from other literature, our finding is particularly of interest as globally, the number of abandoned farmland has been increasing as rural population decreases and many people (particularly young adults) migrate to urban areas (United Nations, 2019). In West Africa, this is no different as the region is currently witnessing a significant shift in its population demography due to a high urban migration trend which is influenced by a multitude of socio-economic factors including conflict, economic opportunities, etc. A land abandonment trend and the maintenance of traditional land-use systems could be a key for the conservation and maintenance of biodiversity. Despite this success, what is not clear is the length of years (recovery time) required for abandoned tropical fallows to reach a biomass state that is equivalent to that of a mature forest. This is highly variable, and is also a source of strong criticisms. For example, a timeline of 200 years was suggested by Saldarriaga et al. (1988) as the recovery period for an area previously utilized for slash-and-burn agriculture whereas Uhl et al. (1982) estimated a period of approximately 100 years for the recovery of a forest ecosystem after cutting and burning. In stark

contrast, Scott (1977) suggested that a period of 30–40 years is sufficient recovery time for forest succession.

In summary, our result emphasize the following pathways: (i) that croplands are not a sustainable land-use as they neither enhance species richness nor support plant diversity. (ii) The potential of abandoned agricultural lands and how much these neglected land-use types will inadvertently support the recovery of biodiversity (iii) that protected areas and fallows in the semi-arid zone do not differ in woody vegetation diversity as, there are no significant differences in species richness and Shannon's diversity for both land-use types. With regards to (iii) above, the implication is that fully protected National Parks do not necessarily conserve the greatest diversity of tree species especially in African dry tropical forests or semi-arid ecosystems. A trend already confirmed by Banda et al. (2006), Paré et al. (2010), and Zisadza-Gandiwa et al. (2013) further indicating that a suite of different types of protection strategies may be the key in the conservation of woody vegetation in this climate zone.

4.2. H1b: Climate effect on Taxonomic diversity

Our second expectation was that taxonomic diversity would successively decrease with an increase in climate conditions. Our results revealed that the number of species increases significantly with the climatic gradient from semi-arid to humid climate conditions and similarly the same was observed for Shannon's diversity index, thus confirming our second expectation that climate plays a fundamental role in shaping the diversity of woody vegetation in West Africa. The reason for these variations in taxonomic diversity may be ascribed to stark differences in environmental conditions between the two zones and a combination of biotic, abiotic and social factors, including predation, competition, seasonality, precipitation, and temperature. Generally, several studies have also shown that tree growth is determined by temperature, moisture, topography, local soil characteristics, among others (Bellefontaine et al. 2000, Yao et al. 2020); factors which are sometimes limiting in a semi-arid savanna ecosystem. Due to these limits, it is assumed that there is a tough competition for resources in these type of ecosystems (Walker et al. 1982) which further initiates a limitation to the growth and abundance of woody vegetation. Studies such as the ones conducted by Bognounou et al. (2009), Sambaré et al. (2010), and Heubes et al. (2013) demonstrated a variation in species richness along a latitudinal climate gradient with a gradual increase in species richness from north to south. Similarly, in a global analysis by Yao et al. (2020), a strong correlation between climate variables and the biodiversity of forest communities was revealed, thus indicating that areas with more moderate climates tend to support higher taxonomic diversity. Rutherford et al. (2013) and Zerbo et al. (2016, 2018) further reported how the species richness of herbaceous plant communities was primarily affected by climate conditions, in particular by precipitation. Precipitation is one of the most important environmental parameters that support primary productivity in woody vegetation (Scholes, 1997). Its effect on species diversity and tree growth has also been well documented in scientific literature (Foley et al. 2003; Parmentier et al.

2007; Zerbo et al. 2016). Furthermore, several analyses have shown precipitation to be the most limiting driver of ecological processes in Africa (IPCC, 2001; Sankaran et al. 2005). All these support the results from our study which shows that an increasing water availability (density and frequency) will likely result in a corresponding increase in species richness and woody vegetation diversity irrespective. The impact of this 'precipitation-diversity hypothesis' is further validated with the fact that land-use systems in the semi-arid zone show a more pronounced seasonality in the distribution of rainfall, including being exposed to several months of drought in comparison to those in the humid zone.

Another important climatic variable we suggest that influenced the contrast in the diversity and abundance of woody vegetation between the two climate zones is temperature. Temperature both in terms of annual mean temperature and seasonal variability exerts a direct influence on the geographical distribution of plant species. Furthermore and through its effect on plant physiological processes and metabolic rates, temperature can drive the loss or gain of plant species. A comprehensive study by Hawkins et al. (2003) which sought to examine the relationship between temperature and diversity highlighted a strong correlation between species richness and temperature in terrestrial ecosystems. Sommer et al. 2010 and Heubes et al. (2013) further showed how an increase in temperature leads to a loss of species, thus explaining the variation in taxonomic diversity that we observed along a latitudinal climate gradient. In addition, the influence of a very high competition for sunlight which is an important mechanism for maintaining species diversity in tropical rain forests (Kobe, 1999) could further explain the reasons for a higher woody vegetation diversity in the humid zone.

Other possible explanations for the observed differences in diversity might be connected to contrasting micro-site factors (e.g. anthropogenic effects) and their variations between both climatic regions; particularly since it has already been established that disturbances from human activities and land-use change are key drivers of taxonomic diversity. The northern region of our study area presents a stark contrast to the south in terms of socioeconomic livelihoods. In the humid zone, agriculture is often characterised with the integration of trees (cocoa, rubber, oil palm) on farms. This is in contrast to the semi-arid zone where, subsistence farming and animal husbandry (rearing of livestock, including cattle and small ruminants) are the predominant agricultural practices. These type of land-use activities play a pivotal role in either supporting or limiting taxonomic diversity. Studies have shown that agro-ecological approaches such as agroforestry could support diversity (Leakey, 2001) while subsistence agriculture and a high dependence on woodlands for incomes could promote the encroachment of agriculture into natural ecosystems which would drive a reduction in taxonomic diversity. We also observed during our field campaigns that croplands in the northern region were often clear-cuts and farmers practiced a rotational cropping system on their farms which perhaps explain why agroforestry practices are not fully employed. Another reason to also consider for the significant differences in woody vegetation between the two climates is the varying management practices. The northern region is characterised with the presence of a very strong fire regime and here also, a higher emphasis is also placed on the protection of fauna due to the tourism potential (in protected areas). Braithwaite (1996) showed how a protection

of the tree layer in savannas from burning could lead to an increase in diversity. Game density (in particular, those of browsing herbivores), is expectedly higher in forested areas which potentially leads to modulations in the biodiversity of savanna ecosystems (O'Connor and Page, 2013) including via a decrease in species richness from overgrazing (Rutherford et al. 2014), an increase in the mortality rates of woody species (O'Connor, 2017; Amahowé et al. 2018) and a limited resprouting ability of juveniles from trampling.

The implication of our results show that in a transition to more arid climatic conditions (a pattern which is attributed to modern day climate change (IPCC, 2014a)), we are likely to have a significant reduction in species richness and general taxonomic diversity across most land-use types (particularly, those with a high intensity of usage).

4.3. H1c: Interactive effect of Land-use and Climate variables on Taxonomic diversity

Our third expectation was that the interaction of land-use change and climate change has a significant effect on Taxonomic diversity i.e., the negative effects of increasing climatic aridity will be more pronounced under an intensive land-use. An interaction occurs when two explanatory variables (in our case: land-use change and climate change) have an effect on a response variable (taxonomic diversity) that is greater or lesser than when these variables act independently. Studies such as the one by Oliver and Morecroft (2014) have shown that these interactions are often times synergetic i.e., that the interaction of these drivers have greater effects than independently. This is somewhat expected as changes in climatic patterns drive changes in how land is used (for example; via changes to socioeconomic systems) and simultaneously, modern day climate change and extreme climate events are strongly influenced by changes in land-use (for example; alterations in carbon dynamics of terrestrial and atmospheric pools as a result of changing land-use patterns (Bonan, 2008)). A novel approach was thus implemented in our study design, by investigating how the interactions that occur between land-use and climatic drivers would impact taxonomic diversity. From the results of our study, we found out that as land-use intensity increases, the effect of climate on taxonomic diversity increases (i.e. the lowest species diversity and richness were found in croplands from the semi-arid climate zone). The direct implication of this is that extreme weather conditions would exacerbate disturbance-induced mortality of woody species. Our result aligns with the modelling study conducted by Heubes et al. (2013) which projected that an interaction of future changes in climate and land-use will significantly reduce woody vegetation diversity in West Africa. We therefore suggest that a continued protection of land-use types such as forests and fallows should be encouraged because these lesser disturbed land-use types are able to minimise the effects of climate change through their contributions to local climatic conditions.

4.4. H2a: Land-use effect on the health of tree populations

From our results, we were able to document that land-use intensity has a strong influence on the health of tree populations as both vegetation structure and population density significantly differed across all land-use types; thus confirming our fourth expectation that the health of tree populations decreases with a corresponding increase in land-use intensity. Croplands in both climate zones had the lowest density of trees while forest had the highest. Simultaneously, they also had the lowest juvenile to adult proportions. The observed differences corroborate our expectation and that of similar studies by Lykke et al. (1998b) and Gouwakinnou et al. (2009) which showed that human disturbances and activities are associated with a lower tree density. We assume that the reduced density of woody vegetation observed particularly in croplands is as a result of high-intensity and high-frequency disturbance regimes including burning, woodcutting, and the clearing of land for agriculture. A pattern similar to the one that has been described by Ouédraogo et al. (2015). Frequent disturbances such as woodcutting and burning hinders the transition from seedling to sapling or to even larger size-classes; a process which often takes a long time. Furthermore, agricultural activities such as repeated burning or extensive weeding can limit stump sprouting, facilitate the depletion of seed banks and increase seedling predation (Nepstad, 1989); or prohibit succession (Uhl and Kauffman, 1990). Incorporating a diversified tree layer as a component of agro-pastoral systems is therefore recommended as a strategy to facilitate the development of a healthy population density in West African croplands. Although, it is important to point out that this recommendation is currently implemented (but only partly) in some croplands, as it was observed during our field campaign that certain trees were left standing on farmlands (and their juveniles even protected) such as *Vitellaria paradoxa* in the semi-arid zone or *Terminalia ivorensis* in the humid zone due to the specific roles these species play especially in soil and water management, shade provision, economic value, or spiritual and cultural reasons. We however consider this management practice to not be sustainable or sufficient enough, particularly for the persistence of tree populations which are not of social or economic importance.

With stem diameter class distributions, we were also able to show that disturbance intensity via land-use change has a strong influence on the population structure of woody vegetation in West Africa. Primarily, four frequency distribution curves have been described when using SCDs to evaluate land-use effects on demographic structure (Lykke, 1998a; Helm and Witkowski, 2012; Ouédraogo et al. 2015): (1) Adult dominated populations with normal (bell-shaped) curves and a positive regression slope; (2) Healthy recruiting populations with a relatively stable size structure (reverse J-shaped curves), with either slightly negative or flat regression slopes; (3) Juvenile dominated populations with comparatively few large individuals (normal curves) and negative regression slopes; or (4) Populations with “missing size classes” (bimodal curves) with negative regression slopes or the dominance of a few size classes. In our study, most land-use types (except croplands) showed a reverse J-shaped curve with more individuals in the lower diameter classes relative to the higher diameter classes; furthermore, the

presence of more individuals in the lower diameter classes subsequently led to significant negative slopes for all land-use types (with the exception of croplands in the humid zone which had a non-significant negative slope). These patterns are a strong indicator that the land-use types of Forest and Fallow do not only have a stable and expanding population but also a recruitment regime that is continuous and sustainable. The occurrence of a natural recruitment regime is a crucial component in forest dynamics due to its strong influence on recovery and the sustenance of forest ecosystem services. Although we do not know the specific drivers that resulted in the patterns of high juvenile densities and a subsequently healthy population in land-use types with low disturbance, we suggest that it is a result of a healthy balance between recruitment and disturbance regimes with healthy populations. A study by Helm and Witkowski (2012) showed how abiotic factors are able to mitigate expected disturbance effects and prevent unstable population structures. In contrast however, we observed that croplands across both climate zones had low juvenile proportions, a somewhat flat SCD slope and in general, expectedly low numbers of both juvenile and adult individuals; an indication that trees in this land-use type across all diameter class are extensively utilized or overharvested. Here, the reasons for this trend are very clear; a high impact of disturbances both in frequency and intensity. The cropland land-use system particularly in West Africa is heavily influenced by human activities and as a result has a very high disturbance regime including for crop cultivation, livestock grazing and firewood harvesting. In addition, agricultural activities can lead to topsoil erosion or an infertile soil which could limit the regeneration of seedlings. Sellan et al. (2019) showed how soil characteristics influenced forest structure; wherein there was a positive correlation between tree structure and soil fertility especially in non-disturbed vegetation areas. Several studies have also shown how the size class distribution of tree species are affected by many factors (Condit et al. 1998) including human disturbances (Bhuyan et al. 2003; Schumann et al. 2010) or protection effect (Fandohan et al. 2011; Houeohanou et al. 2013; Ouédraogo et al. 2015). Luoga et al. 2002 further showed how the harvesting of trees significantly affects the structure of woodlands. This is done via a number of mechanisms including by limiting the sprouting ability of species which can lead to a significant decline in population (Schumann et al. 2011) or by influencing the structure and vital rates (growth, reproduction and survival rate) of vegetation through selective harvesting for timber and for firewood (Muvengwi et al. 2020), a common practice in croplands which in turn can change population demographics (dynamics and structures). Thus, we suggest that the effect of protection and non-protection between Forest and Cropland areas is a major factor driving the observed differences in tree population density and vegetation structure. Our results are similar to the findings by Ouédraogo et al. 2015 which showed that protection is a very crucial tool in ensuring sufficient recruitment of woody vegetation, particularly since protected areas were established for the primary purpose of protecting ecosystems by reducing human land-use activities within their borders.

We also suggest that there could be other disturbance effects responsible for the variation in population density and vegetation structure between the different land-use types including: (1) herbivory and game

management: Large herbivores are often times referred to as landscape architects due to the role they play in the structural management of woody vegetation. Since protected areas support the maintenance of viable populations of animals, they often harbour large populations of both large and small generalist herbivores such as Elephants and Deers which browse on tree recruits and subsequently modulate the spatial structure of ecosystems (Teren et al. 2018; Sandhage-Hofmann et al. 2021) or significantly affect size class distribution of trees (Peltzer et al. 2014). However, what is not explicitly clear is how the effects of these herbivores influence vegetation structure and the health of tree populations as there are contrasting reports. While some studies show a positive feedback; by facilitating seedling recruitment and aiding tree seed dispersal (Brown and Archer, 1999), others have shown a negative feedback; by either suppressing tree density (Staver et al. 2009), reducing regeneration or from overgrazing (Moleele and Perkins, 1998). (2) Fire as a management tool: Fire is an important structural driver, particularly in savanna landscapes (Higgins et al. 2007) where it is often applied as a tool to either prevent a more destructive late fire or to reduce biomass and open up the vegetation cover for an increased visibility. These two factors are beneficial for the germination and growth of seedlings/saplings and in addition, reducing the competitive effects for light and nutrients. Similar to herbivory management however, the role of fire in influencing vegetation structure and health of tree populations is not explicit because depending on the intensity and frequency, fire could act as a tool to negatively affect seed survival and sapling recruitment (Zida et al. 2007) and prevent the establishment of adult trees by killing tree saplings (Higgins et al. 2000) or in contrast can facilitate the establishment of tree seedling (Bond and Van Wilgen 1996).

4.5. H2b: Interactive effect of Climate and Land-use on the health of tree populations

Temperature and precipitation control many ecological and physiological processes including nutrient cycling, tree growth and size which can ultimately influence tree size class structure (Coomes and Allen, 2007). Thus, we expected that an interaction of changing climate conditions and increasing land-use intensity would significantly affect the health of tree populations. More specifically, we expected that the negative effects of land-use intensification on population health would be more pronounced under more arid conditions i.e. the slopes of croplands in the semi-arid zone would be positive and significantly different from the slopes of other land-use types. However, our results showed that the interaction of climate and land-use had no significant effect on the health of tree populations ($F = 0.142$, $P > 0.05$). From the results of our study, we therefore conclude that disturbances related to land-use are the most significant factor influencing the growing conditions of woody vegetation and their structure in West Africa (Brandt et al. 2014).

5. CONCLUSION

Understanding the relationships between biodiversity and global change drivers is a crucial point to facilitate effective management decisions. In general, our results show that increasing land-use intensity and changing climatic conditions has a considerable effect on the structure and diversity of woody vegetation in West Africa. Whilst changes in land-use affects the spatial heterogeneity of woody vegetation, changing climate conditions affects its distribution and abundance. Our study which is based on an extensive field data and the exploitation of natural gradients in climate and land-use shows that patterns of species richness and woody are strongly influenced by not only specific drivers such as climate change and human activities but also by an interaction of these drivers. This is particularly important since often times, it is the case that multiple drivers act together to impact diversity. It is important to point out that although our study investigated two of the main drivers of global biodiversity loss, there are other drivers worth considering (such as the threat of invasive species and pollution) that might also interact with land-use change and climate change to impact biodiversity; thus, suggesting potential areas for future research. As a recommendation, our study also strengthens the emphasis that is currently being placed on fallow lands/abandoned croplands as a biodiversity hotspot and a potential pathway for the recovery of plant diversity. It is pertinent to note however that this cannot be replaced as an active restoration process since the pathway of forest succession is influenced by many other factors. In the case of vegetation structure and the health of tree populations in West Africa, the reverse J-shaped size class distribution curve and significant negative slopes observed in our study indicate that most of the land-use types are in a healthy state. However, consideration should be given to management practices such as the incorporation of a diversified tree layer in croplands so as to promote a higher density of tree population.

In summary, our study adds to existing knowledge within the broad field of global change ecology, particularly in the West African region and also provides a new approach to vegetation studies that are conducted in the region. Regional studies such as this can help in documenting local patterns of spatial and temporal variation in diversity. Knowledge about the patterns of current species distribution can also be used to model/estimate their future distribution. It additionally sheds light on the potential impacts of land-use change and climate change on West Africa's woodlands and provides important insights for policymakers, conservation practitioners, and field researchers working on ecological and environmental challenges. Finally, it contributes to our understanding of how these ecosystems may respond to future environmental changes and informs strategies for their sustainable management and conservation.

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APPENDIX

Formula table 1: showing the different mathematical and statistical equations that were used in the study

Parameter	Formula
1. Relative Abundance (%)	$[\text{total aboveground biomass of a species} \div \text{aboveground biomass of all species (G)}] \times 100$
2. Aboveground biomass (in kg) based on Chave et al. (2005)	$AGB = 0.0673 * (SWD * DBH^2 * h)^{0.976}$, where AGB is aboveground biomass of a tree, SWD is the specific wood density (in g cm ⁻³), DBH is the stem diameter at breast height (in cm), and h is the total tree height (in m).
3. Aboveground biomass (in kg) based on Mbow et al. (2013)	$AGB = 25.581h - 7.006h^2 + 0.662h^3$, where BM is aboveground biomass of a tree and h is the total tree height (in m)
4. Specific wood density (SWD). Calculated	$SWD = \text{Dry mass} \div \text{Volume} (\pi r^2 h)$. For volume ($\pi r^2 h$) where r is diameter readings divided by 2 and h is length of the core)
5. Relative Density (%)	$[\text{number of individuals of a species (n ha}^{-1}) \div \text{total number of individuals (N ha}^{-1})] \times 100$
6. Relative Frequency (%)	$[\text{frequency of species (n)} \div \text{total number of species (N)}] \times 100$
7. Species richness	Based on the total number of species
8. Species evenness (J) =	$H' \div \log(i)$ where H' is the Shannon diversity and I is the total number of species
9. Shannon diversity (H')	$-\sum P_i \ln (P_i)$ where P_i is the ratio of the number of individuals of species i and the total number of individuals
10. Upscaling factor	- Plot level: 5 x 5m (25m²) to 1 hectare (10,000m²) = no. of individuals x 400 - Plot level: 10 x 10m (100m²) to 1 hectare (10,000m²) = no. of individuals x 100 - Plot level: 50 x 20m (1,000m²) to 1 hectare (10,000m²) = no. of individuals x 10 - Plot level: 50 x 50m (2500m²) to 1 hectare (10,000m²) = no. of individuals x 4

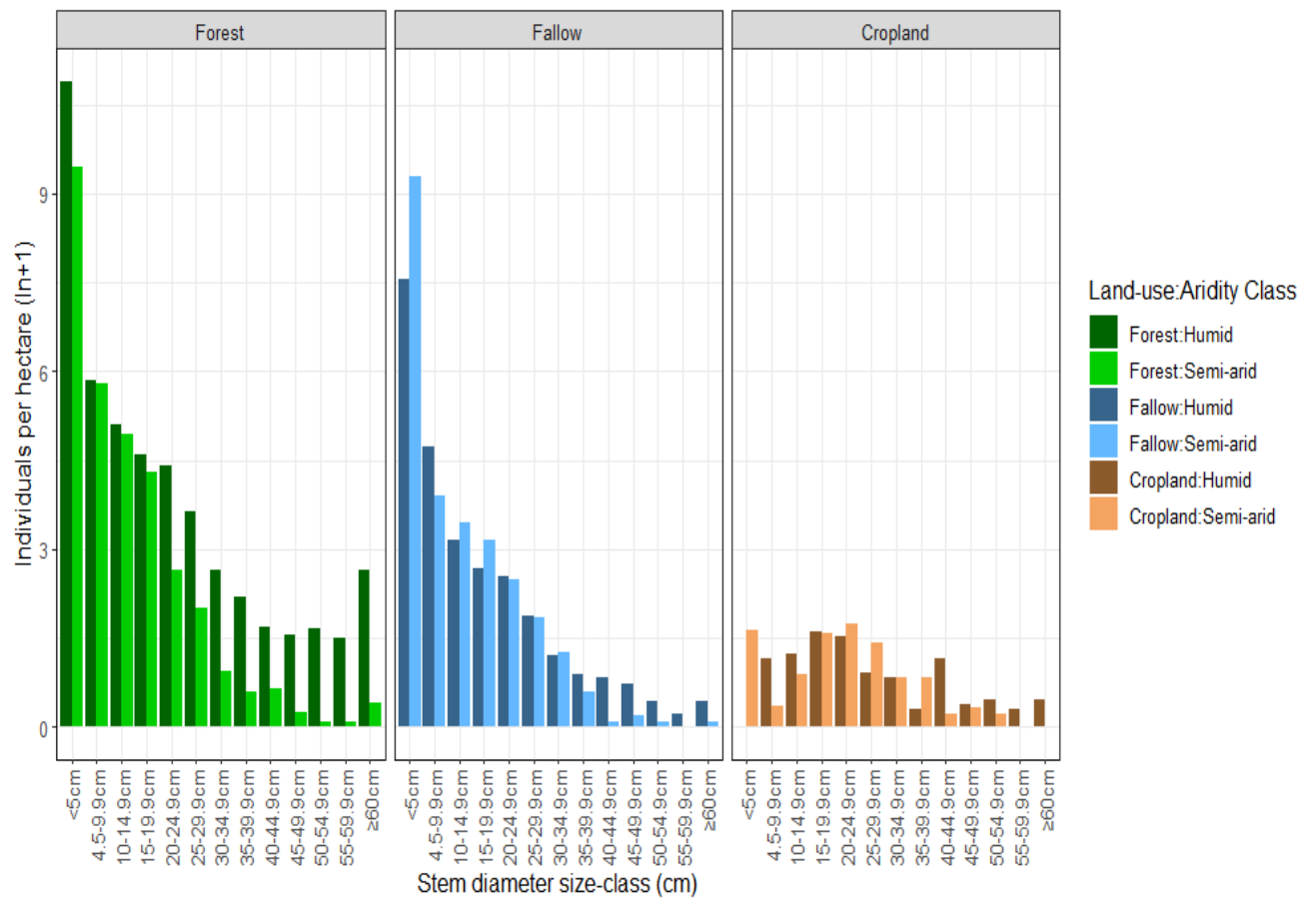


Figure 9: Stem diameter distributions of different land-use types (Forest, Fallow and Cropland) from the humid and semi-arid climate zones of Ghana.

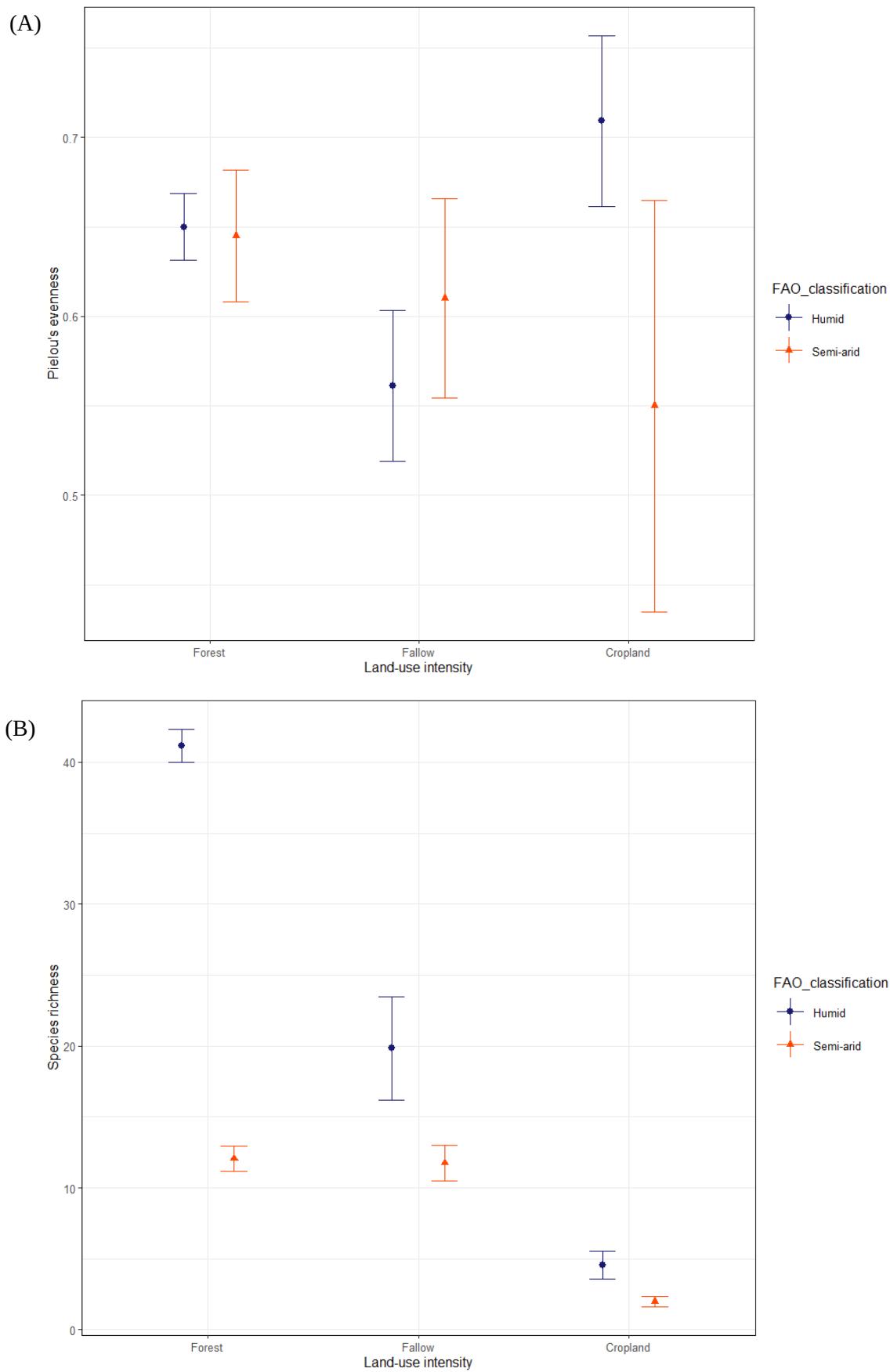


Figure 10: Line plots showing the mean values of various taxonomic diversity indices [(A) Pielou's Evenness, (B) Species richness] along a land-use intensification gradient from the humid and semi-arid climate zones of Ghana. The standard deviation is the upper error bar while standard error is the lower error bar

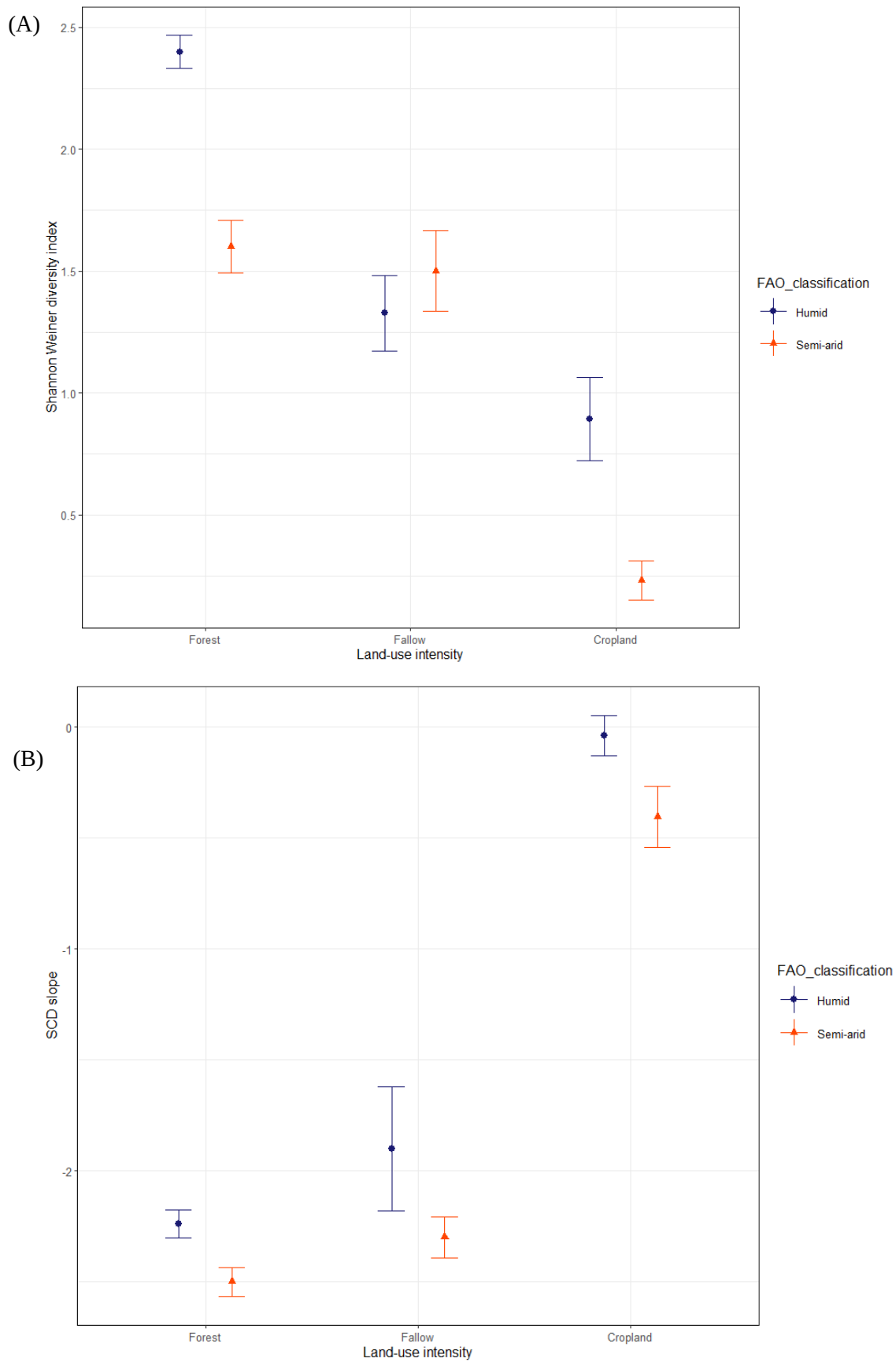


Figure 11: Line plots showing the mean values of Shannon-Weiner's diversity index (A) and Slopes of size-class distribution (B) along a land-use intensification gradient from the humid and semi-arid climate zones of Ghana. The standard deviation is the upper error bar while standard error is the lower error bar